

Voltage quality and economic activity: On-line Appendix

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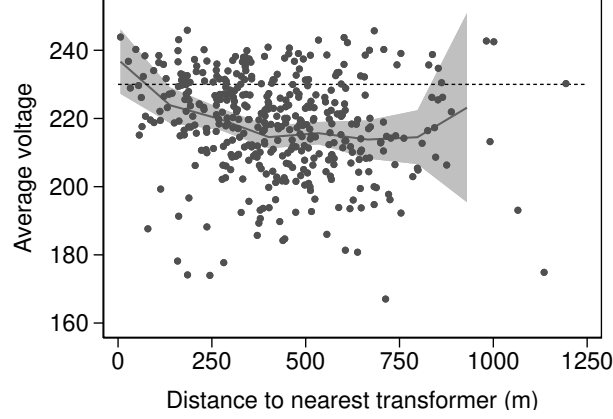
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Latest version available [here](#).

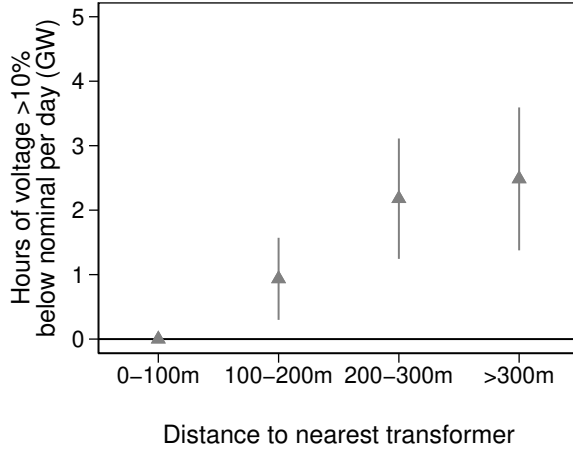
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A Appendix Figures

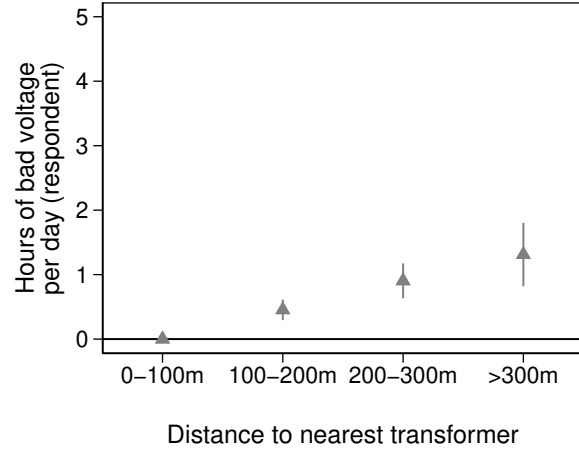
Figure A1: Correlation between average voltage and distance to nearest transformer
Panel (A) Raw correlation with GridWatch devices



Panel (B) GridWatch devices

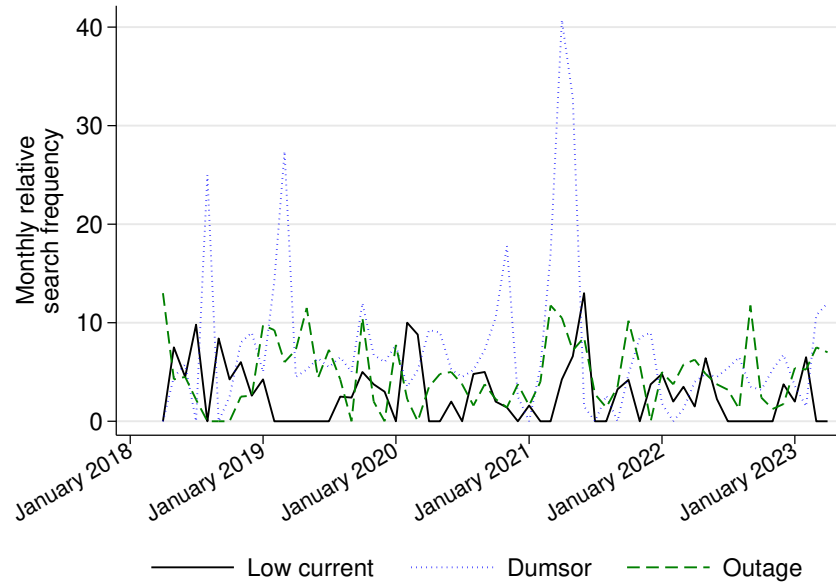


Panel (C) Self-reported



Panel A shows device-level average voltage by the distance along the electricity network from that device to the nearest transformer using GridWatch data from Ghana (described in [Section 3](#)). The solid fitted line shows the best fit from a local polynomial, and the shading show a 95% confidence band. For Panel B, respondents are matched with GridWatch data from the device that is nearest to their location (with “bad” voltage being voltage more than 10% below nominal). For Panel C, respondents are asked about average daily “bad” voltage hours over the 30 days preceding the survey.

Figure A2: Google trends for search on voltage and outage issues in Ghana, April 2018-2023



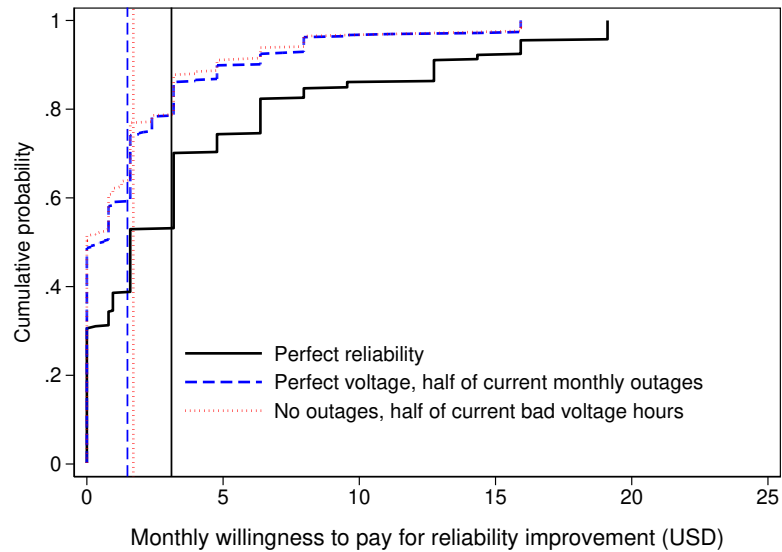
Data are aggregated from weekly Google Trends data for the specified search terms. Relative search frequency is calibrated to the maximum search interest across the three terms over the time period. “Low current” is the most common phrase used in Ghana to refer to issues related to voltage. “Dumsor”, meaning “off and on” in Akan, is a common term to refer to outages in Ghana, and is particularly associated with periods of load shedding and frequent long-lasting outages.

Figure A3: A GridWatch device



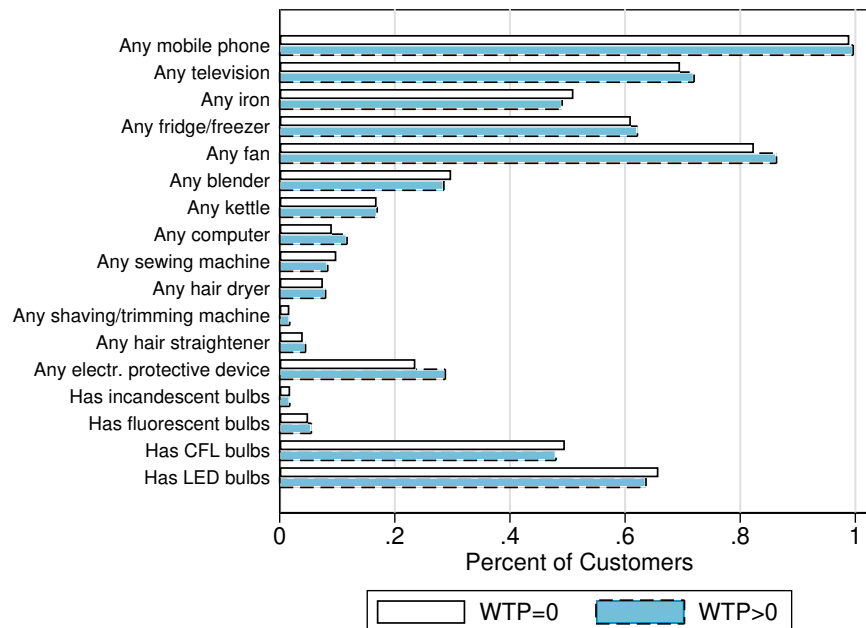
A GridWatch device, part of nLine’s GridWatch technologies used to measure power outages and voltage. Each GridWatch device measures voltage in real-time, stores this on a local SD card, and sends the data to the cloud via a sim card whenever local cellular service permits. A back-end computing technology aggregates these data in real-time, monitoring voltage at the device level and detecting spatial and temporal correlations in power loss and restoration signals to identify power outages with relatively high confidence.

Figure A4: Willingness to pay for electricity connections with particular characteristics



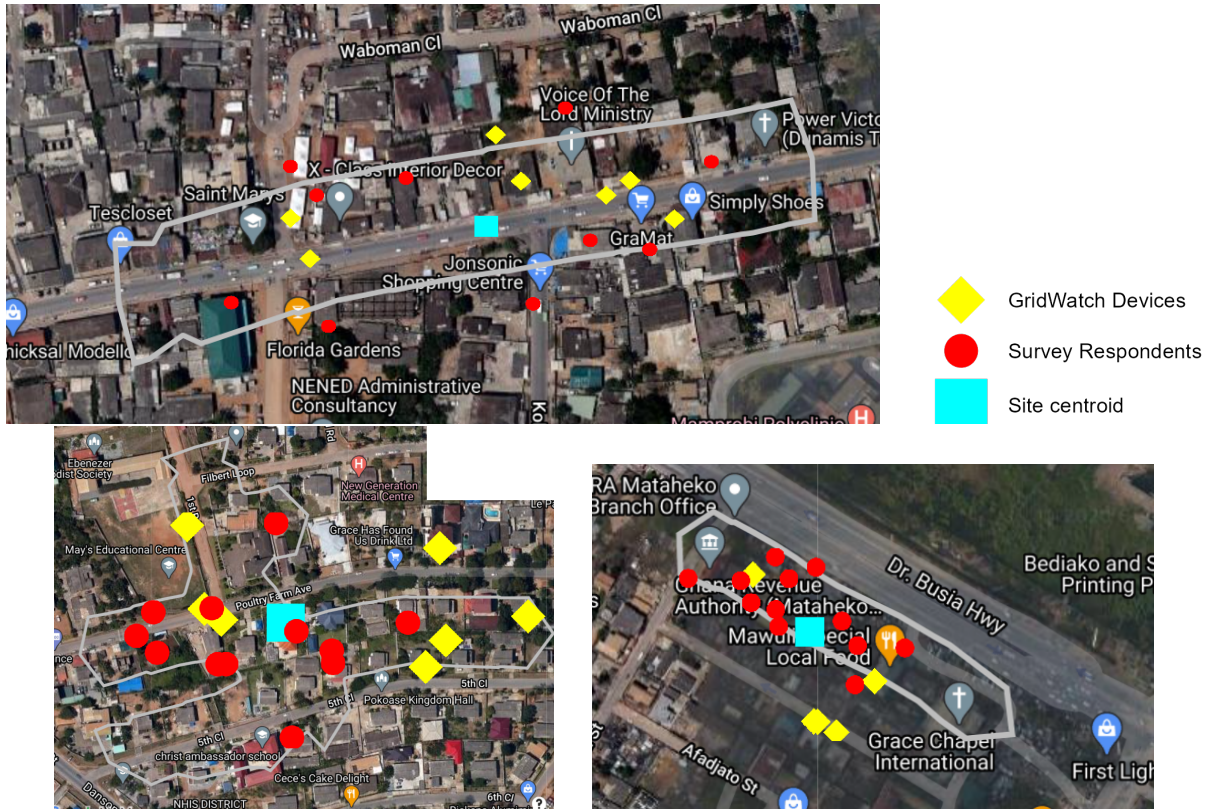
Willingness to pay is elicited as the maximum monthly amount respondents would be willing to pay for an improved electricity connection, in addition to the cost of their electricity consumption. Respondents are asked first about a connection with perfectly reliable electricity and then elicited about connections with specific reliability improvements. Vertical lines indicate the mean willingness to pay for each type of improved electricity connection. The mean monthly electricity spending for both businesses and households is USD 18.

Figure A5: Appliance ownership by willingness to pay for voltage improvements



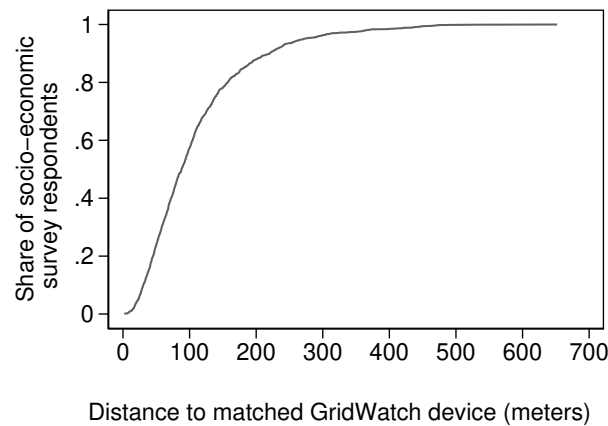
Appliance ownership is at the time of the baseline survey. Willingness to pay is for a connection with half of current outage hours but perfect voltage quality. Results are similar using WTP for a connection with half of current bad voltage hours but no outages.

Figure A6: Example site



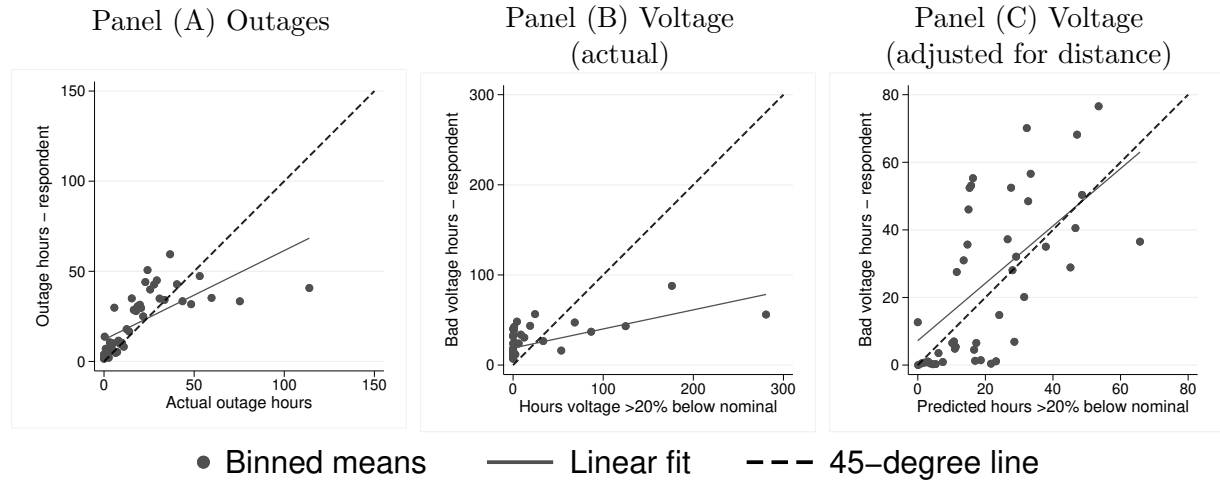
Sample site maps where field officers conducted surveying. The site centroid is where the transformer is injected at treatment sites. The surveying area (outlined in white) is generated by following the LV lines for up to 200 meters from the centroid and then including any area that is within 50 meters of a line segment. To preserve anonymity the locations of devices and respondents have been jittered with a small amount of noise.

Figure A7: CDF of distances



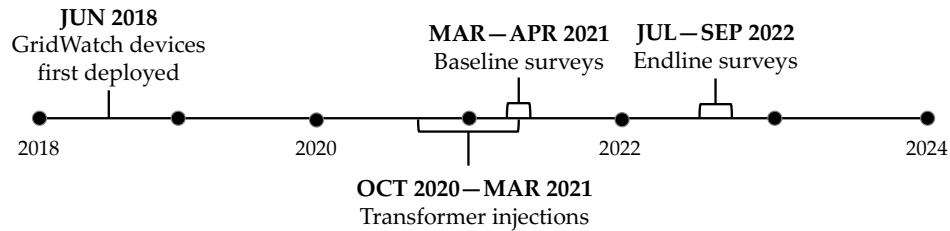
Distribution of distances from each survey respondent to the device whose data they are matched to. The median distance is 88 meters. 80% of the respondents are within 157 meters of their matched device.

Figure A8: Correlations between measured and reported electricity characteristics



Respondents are matched with data from the device that is nearest to their location (horizontal axes). Predicted hours of bad voltage (panel C) are calculated by adjusting data by respondent and device distance to the nearest transformer. Observations are binned into 50 quantiles of approximately 62 observations each.

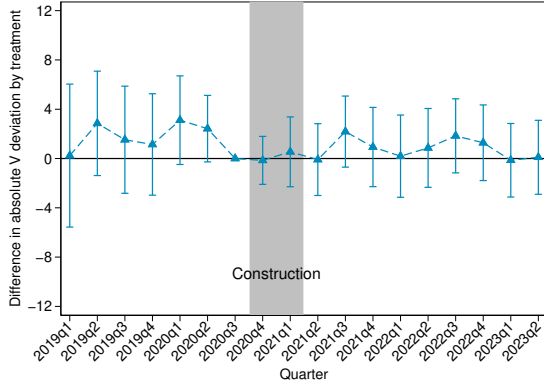
Figure A9: Project timeline



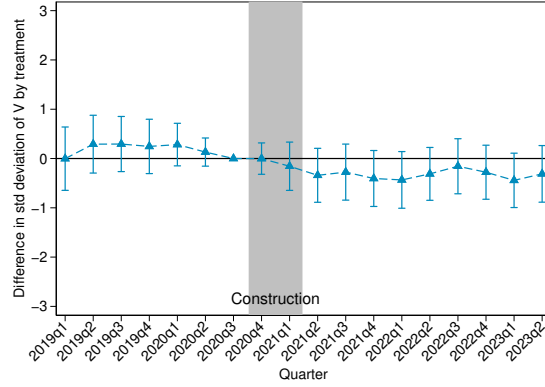
Timeline of research and construction activities. GridWatch data were collected continuously in the months after device deployment. The analyses in this paper include GridWatch data collected through April 2023.

Figure A10: Impacts of transformer injection intervention on power quality and reliability

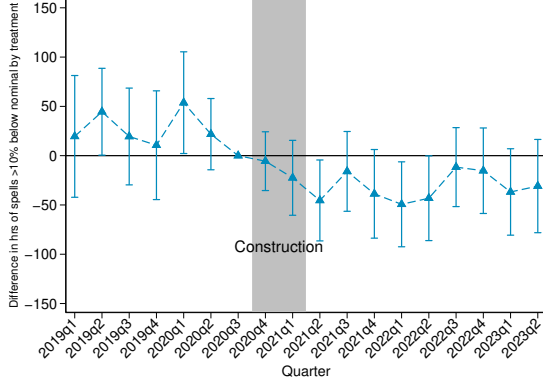
A) Treatment effect on absolute voltage deviation



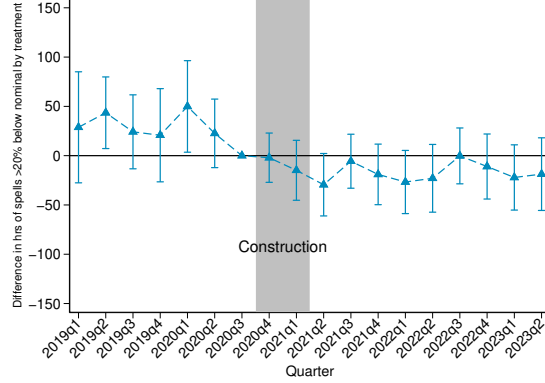
B) Treatment effect on voltage standard deviation



C) Treatment effect on hours of spells >10% below nominal

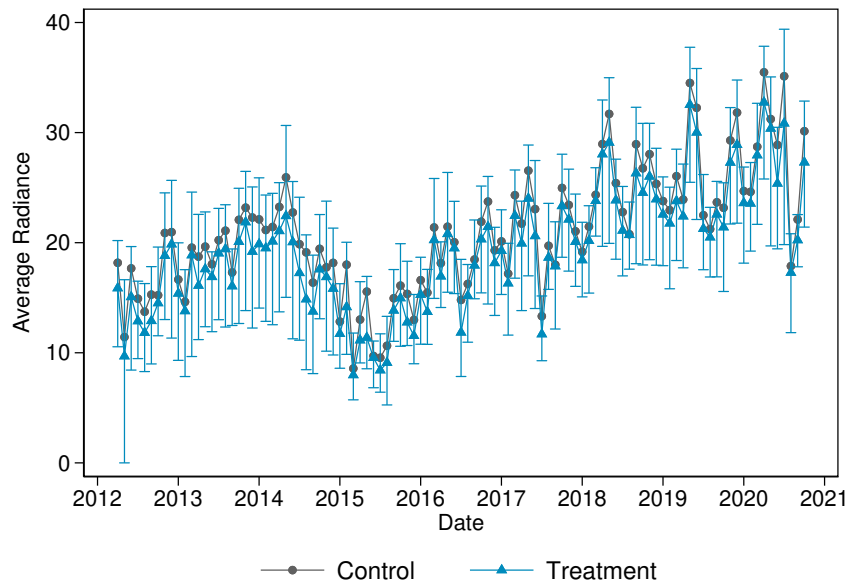


D) Treatment effect on hours of spells >20% below nominal



All panels show estimated coefficients and 95% confidence intervals from a regression of outcomes on site treatment status by quarter. Panels (A) and (B) are estimated using hourly data, controlling for hour of day, week of year, and site fixed effects. In Panel (A), absolute voltage deviation is the absolute value of the hourly voltage deviation from nominal voltage (230 V). Panels (C) and (D) are estimated using monthly data, controlling for month and site fixed effects. SEs clustered by site. Treatment coefficients use $t = -1$ as the reference period and thus do not exactly match those presented in [Table 5](#), which use the entire pre-period as the reference period.

Figure A11: Balance in nightlight radiance

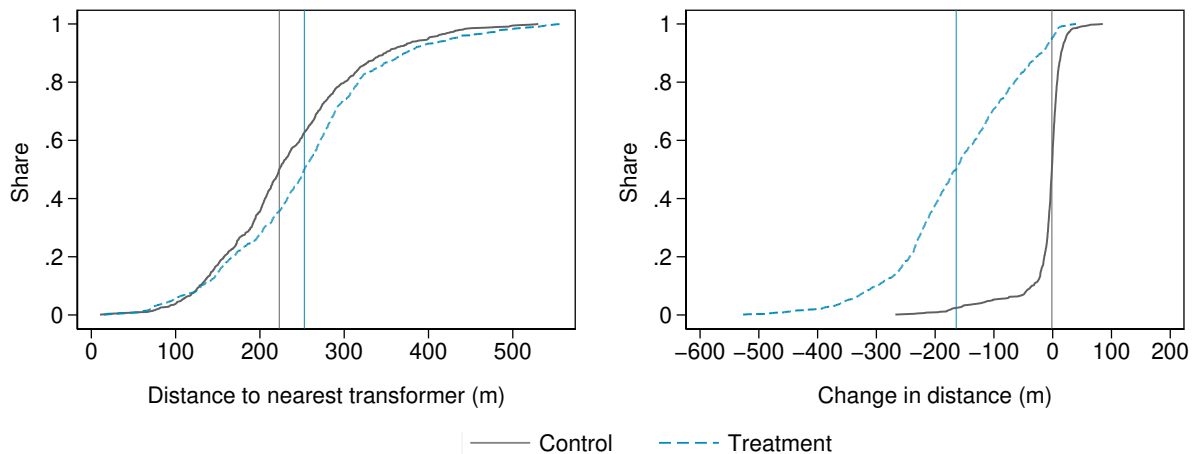


Median monthly nighttime Visible Infrared Imaging Radiometer Suite (VIIRS) radiance between 2012-2020 per site-month, with bands showing the 25th to 75th percentile (Elvidge et al., 2017).

Figure A12: Distance between each respondent and their nearest transformer

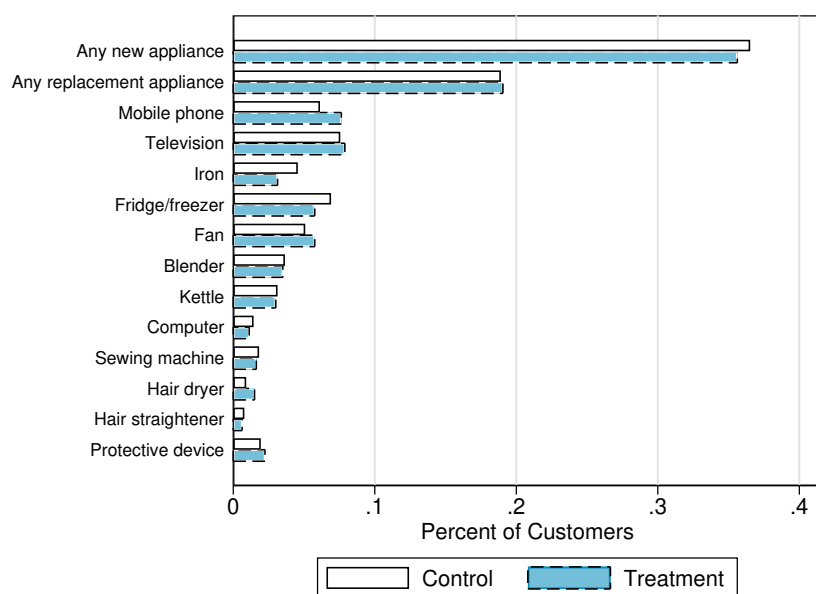
(A) Before construction

(B) Change from before to after construction



Panel A shows the distance (in meters) from each respondent to the nearest transformer at baseline for survey respondents. Panel B shows the change in this distance from baseline to endline. The figures includes a small number of individuals who moved within the survey sites between baseline and endline, which accounts for nearly all the variation in distances to transformer in control sites. Vertical lines mark the median respondents in control and treatment sites.

Figure A13: Appliance acquisition between surveys by site treatment status



Shares of respondents (pooling business and household respondents) who report acquiring that appliance between baseline and endline, across treatment and control sites.

B Appendix Tables

Table B1: Baseline correlates of power quality and reliability

	N	Mean (SD)	Monthly hours > 10% below nominal	Monthly outage hours
Distance to nearest transformer (10m)	1566	24.24 (9.15)	2.17** (0.84)	0.34** (0.17)
Shares electricity meter with other users (=1)	1566	0.40 (0.49)	-0.69 (7.92)	-1.53 (1.70)
Wealth index (normalized)	1566	-0.14 (0.98)	-1.44 (4.83)	-0.37 (1.26)
Household members	741	3.63 (1.91)	1.14 (2.44)	0.45 (0.62)
Monthly household income (USD 100s)	709	3.36 (4.86)	-0.81 (1.13)	-0.04 (0.20)
Number of workers	825	1.95 (1.99)	-1.81 (1.89)	-0.20 (0.29)
Last month business revenue (USD 100s)	719	3.87 (5.68)	-1.37* (0.75)	0.16 (0.16)
Outcome mean			128.0	25.5

The table presents results from 14 regressions, with two baseline measures of power quality from GridWatch devices deployed near survey respondents as the outcome variables (columns) and 7 survey respondent characteristics as the independent variables (rows). Voltage data is not available for respondents in one site where no devices were deployed. All regressions control for respondent sex, age, type (household or business), and rental or ownership status. The wealth index includes roof and wall material quality, count of owned appliance types, and secondary education completion. SEs clustered by site. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B2: Baseline balance by site status

	Control N	Mean	Treat N	Difference	p-val	Accra Mean
<i>Respondent and Location</i>						
Age (years)	772	38.79	803	-0.85	0.141	45.39 ^a
Respondent is male (=1)	772	0.36	803	0.01	0.753	0.67 ^a
Completed secondary education (=1)	772	0.50	803	-0.00	0.983	
Owns premises (=1)	772	0.37	803	-0.02	0.368	0.59 ^a
<i>Appliances</i>						
Any television (TV) at location (=1)	772	0.68	803	-0.05	0.040	0.85 ^b
Any fridge at location (=1)	772	0.63	803	0.02	0.434	0.62 ^b
Count of mobiles	772	2.23	803	0.13	0.107	3.02 ^a
Any reliability protective device owned (=1)	772	0.25	803	-0.02	0.461	
Count of reliability protective devices	772	0.35	803	0.01	0.694	
<i>Electricity and Energy</i>						
Pays someone else for electricity (=1)	772	0.09	803	0.01	0.611	
Count of meter users	772	1.76	803	-0.11	0.141	
Last month electricity spending (USD)	763	17.59	796	1.78	0.026	15.34 ^a
Has generator (=1)	772	0.04	803	0.00	0.968	0.02 ^a
Count of alternative fuels used in past 3 months	772	0.92	803	0.00	0.919	
Last month spending on alternative fuels (USD)	772	7.93	803	1.27	0.241	
<i>Reliability</i>						
Reported number of outages in past month	772	6.98	803	0.35	0.139	
Reported total outage hours in past month	772	38.61	803	-1.47	0.541	
Reported avg. hours per day with bad voltage	769	1.44	797	-0.29	0.062	
Any appliance damaged by voltage in past year (=1)	772	0.25	803	-0.04	0.056	
Amt. spent on burnt/broken apps in past year (USD)	768	9.18	794	-0.25	0.877	
<i>Household Characteristics</i>						
Adult members	363	2.38	383	-0.03	0.758	2.11 ^a
Child members (<18)	363	1.19	383	-0.04	0.721	1.34 ^a
Last month HH income (USD)	347	330.63	367	-7.06	0.846	328.25 ^a
Share of HH adults (18+) with paid jobs in last 7 days	363	0.64	383	-0.03	0.265	
<i>Business Characteristics</i>						
Number of workers	409	1.99	420	0.04	0.790	7 ^c
Last month business revenue (USD)	343	397.94	380	-0.54	0.990	7187.50 ^c
Last month business costs (USD)	325	283.14	366	-25.99	0.359	
Last month business profit (USD)	310	102.40	336	8.18	0.457	1851.44 ^c
Usual business open hours	409	12.16	420	0.19	0.268	
Any non-electric business machines at location (=1)	409	0.09	420	-0.00	0.905	
Business engaged in retail activities (=1)	409	0.44	420	-0.00	0.933	
Business engaged in manufacturing activities (=1)	409	0.22	420	0.02	0.378	
Business engaged in other service activities (=1)	409	0.35	420	-0.02	0.514	
Business activity likely using electricity (=1)	409	0.23	420	0.01	0.710	

This table shows means in the baseline period for survey respondents, pooling businesses and households, and tests for significance of the differences in means by line bifurcation treatment status. The p-value for the joint F-test for household baseline characteristics is 0.230. The p-value for the joint F-test for business baseline characteristics is 0.407. Summary statistics for the population of households in Accra are taken from Ghana Statistical Service data from the 2017 Ghana Living Standards Survey (^a) or the 2015 Labor Force Survey (^b) for urban households in the Greater Accra Region and calculated using survey weights to generate representative estimates. Summary statistics for the population of businesses in Accra are taken from Ghana Statistical Service data from the 2015 Integrated Business Establishment Survey II (^c) for businesses in urban Accra with 30 or fewer employees, which are sampled randomly from the 2013 census of Ghanaian businesses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B3: Correlations between voltage quality and primary outcomes

	Full N	Mean (SD)	Adjusted average voltage	Adjusted hours voltage 20% below nominal
Reported hours of bad voltage in past month	3037	23.02 (60.61)	-0.732*** (0.198)	0.161*** (0.052)
Reported total outage hours in past month	3048	20.50 (34.13)	-0.298*** (0.091)	0.044*** (0.017)
WTP for perfect reliability and quality (USD)	3057	2.50 (3.46)	-0.019*** (0.005)	0.003*** (0.001)
WTP for perfect voltage and half outage hours (USD)	3057	1.31 (2.13)	-0.010*** (0.003)	0.002** (0.001)
WTP for no outages and half bad voltage hours (USD)	825	1.89 (3.11)	-0.005 (0.008)	0.002 (0.002)
Voltage damage and protection index	3057	-0.02 (0.97)	-0.003* (0.002)	0.001** (0.001)
Any voltage-related damage, last 12 months (=1)	3057	0.24 (0.42)	-0.002** (0.001)	0.001** (0.000)
Amt. spent on burnt/broken apps in past year (USD)	2989	6.56 (22.94)	-0.083*** (0.029)	0.017** (0.008)
Amt. spent on burnt/broken apps (if damage = 1)	692	30.36 (47.69)	-0.089 (0.134)	0.004 (0.024)
Any reliability protective device owned (=1)	3057	0.25 (0.43)	-0.000 (0.001)	0.000 (0.000)
Value of protective devices owned (USD)	2587	4.08 (14.51)	-0.035 (0.022)	0.013* (0.007)
Any alt. energy source used in last month (=1)	3057	0.05 (0.21)	0.000 (0.000)	0.000 (0.000)
Total no. of appliances owned	3057	8.48 (5.38)	0.003 (0.006)	0.000 (0.001)
Last month electricity spending (USD)	2959	14.96 (12.87)	0.010 (0.016)	-0.003 (0.003)
Last month business profit (USD)	1079	88.93 (127.63)	0.058 (0.226)	0.021 (0.045)
Last month business revenue (USD)	1245	392.85 (574.60)	1.873** (0.901)	-0.355* (0.213)
Last month business costs (USD)	1179	292.65 (386.76)	1.215* (0.651)	-0.171 (0.146)
Last month HH income (USD)	1314	301.84 (430.75)	-0.715 (0.748)	0.176 (0.158)

This table shows a different regression in each cell, with two voltage measurements as the explanatory variables and each row a different outcome variable. Each row represents a different outcome pooling business and household respondents. Profit is measured by directly asking the respondent, rather than by subtracting costs from revenues. Total reported costs are the sum of costs for specific items/activities and are not comprehensive. The columns indicate measures of voltage quality—the independent variables. Voltage is measured by assigning each respondent GridWatch data based on the nearest devices for the year preceding the survey date. Voltage measures are adjusted to account for differences between the distance to the nearest transformer for the respondent and their matched device. Mean voltage in control sites is 219.5V at baseline and 224.6V at endline. In all the regressions, we also control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the location includes both a household and a business, and district fixed effects. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ [Table C10](#) shows the same using raw matched device data.

Table B4: Testing voltage spillovers in control sites from transformer injection intervention

	Average voltage	
	(1)	(2)
Post construction=1	7.37*** (2.00)	6.53* (3.44)
Post construction=1 $\times$ Below median distance to nearest injection site=1	-3.79 (2.90)	
Post construction=1 $\times$ Distance to nearest injection site (100m)		-0.06 (0.17)
Observations	4936545	4936545
Pre-construction mean, above median distance to injection	220.09	220.09
Hour of day FE	Y	Y
Week of year FE	Y	Y
Site FE	Y	Y

This table tests for differences in how voltage changed in control sites—which did not receive any new transformers—after the transformer construction intervention by distance along the grid network from the control site to the nearest new injection transformer. The outcome is average voltage, measured using hourly voltage data at the GridWatch device level. Column (1) tests for differences by whether a device is in a site below the median distance to the nearest injection transformer, while Column (2) tests for differences by distance, measured in 100m. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B5: Impact of transformer injection intervention on customer electricity experience

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post $\times$ Treat (SE)
Voltage damage and protection index	3150	0.00 (1.00)	-0.11** (0.05)	0.09 (0.06)	-0.10* (0.06)
Any voltage-related damage, last 12 months (=1)	3150	0.25 (0.43)	-0.05* (0.03)	0.04 (0.03)	-0.05 (0.04)
Any reliability protective device owned (=1)	3150	0.25 (0.44)	-0.02** (0.01)	0.02 (0.02)	-0.02 (0.01)
Amt. spent on burnt/broken apps in past year (USD)	3080	9.28 (33.80)	-5.77*** (1.37)	0.05 (1.77)	1.23 (1.91)
Value of protective devices owned (USD)	2668	5.49 (23.16)	-2.32** (0.92)	0.36 (1.42)	0.32 (1.63)
Reported hours of bad voltage in past month	3110	43.18 (87.51)	-42.48*** (4.72)	7.60 (7.41)	-9.09 (7.70)
Reported total outage hours in past month	3081	32.15 (31.08)	-29.32*** (2.05)	1.69 (2.60)	-1.18 (2.74)
WTP for perfect reliability and quality (USD)	3150	3.29 (4.41)	-1.41*** (0.21)	-0.34 (0.25)	0.44 (0.28)
WTP for perfect voltage and half outage hours (USD)	3150	1.58 (2.71)	-0.37** (0.16)	-0.17 (0.17)	0.21 (0.21)
WTP for no outages and half bad voltage hours (USD)	850	1.95 (3.34)	-0.07 (0.30)	-0.37 (0.32)	0.99** (0.44)
Total no. of appliances owned	3150	8.55 (5.99)	-0.02 (0.09)	-0.07 (0.35)	0.09 (0.13)
Plans to purchase ≥ 1 new appliance in next year	3150	0.58 (0.49)	-0.05* (0.03)	-0.02 (0.03)	0.04 (0.04)
Any alt. energy source used in last month (=1)	3150	0.05 (0.22)	-0.01 (0.01)	0.01 (0.01)	-0.00 (0.01)
Last month electricity spending (USD)	3050	17.72 (16.95)	-3.92*** (0.60)	-1.90* (1.02)	0.53 (0.80)
Last month business profit (USD)	1104	98.60 (143.83)	-6.77 (10.50)	-12.44 (11.53)	3.55 (13.67)
Last month business revenue (USD)	1280	396.21 (625.91)	84.40* (44.75)	-0.80 (51.71)	-92.58 (57.75)
Last month business costs (USD)	1206	276.34 (358.93)	82.04** (36.68)	25.30 (37.84)	-98.48** (49.45)
Last month HH income (USD)	1358	332.33 (473.05)	-34.24 (36.83)	16.93 (41.60)	-75.43 (49.79)

This table shows the difference-in-difference results from the [Equation 2](#) pooling businesses and households. Each row presents results from one regression with a different socio-economic variable as the outcome. All outcomes pre-specified in the pre-analysis plan (Berkouwer et al., 2019), except for voltage improvements as these were unanticipated. All variables measuring values are in USD. Results are qualitatively unchanged when using logged versions of continuous outcomes. Sample sizes vary for some questions because of missing data, particularly when respondents were unable to estimate monetary values with a high degree of confidence, or because some questions were only asked to a subset of respondents. Reliability outcomes are measured using respondent self-reports based on the 30 days prior to the survey date at both baseline and endline. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is a household or a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Sharpened FDR q-values following Anderson (2008) are shown in [Table C8](#). All effects of Post remain statistically significant after this adjustment, but the significant effects of Post $\times$ Treat do not.

Table B6: Impact of transformer injection intervention on business outcomes

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post $\times$ Treat (SE)
Last month business profit (USD)	1104	98.60 (143.83)	-6.77 (10.50)	-12.44 (11.53)	3.55 (13.67)
Last month business costs (USD)	1206	276.34 (358.93)	82.04** (36.68)	25.30 (37.84)	-98.48** (49.45)
Last month wage and benefits costs (USD)	1330	54.03 (135.87)	14.97* (8.18)	-5.78 (10.38)	-5.75 (11.16)
Last month materials costs (USD)	1266	187.49 (297.45)	72.19** (29.95)	47.78* (27.96)	-93.21** (40.69)
Last month electricity spending (USD)	1594	17.63 (16.99)	-3.92*** (0.73)	-1.74 (1.22)	0.21 (0.96)
Last month spending on alternative fuels (USD)	1658	5.14 (37.42)	-0.95 (1.75)	-1.38 (2.05)	1.17 (1.90)
Last month business revenue (USD)	1280	396.21 (625.91)	84.40* (44.75)	-0.80 (51.71)	-92.58 (57.75)
Estimated increase in revenue w/ perfect electricity (USD)	1044	545.75 (2038.28)	-355.67*** (137.04)	-125.66 (149.10)	30.66 (156.00)
Number of workers	1658	1.99 (1.90)	0.11* (0.06)	-0.06 (0.14)	0.07 (0.09)
Share of men employees	1646	0.31 (0.42)	-0.01 (0.01)	-0.00 (0.01)	-0.00 (0.02)
Share of full-time employees	1628	0.91 (0.21)	-0.05*** (0.02)	0.01 (0.02)	0.00 (0.02)
Business open during any “dark” hours	1658	0.77 (0.42)	-0.08*** (0.03)	-0.01 (0.03)	-0.02 (0.04)
Total hours typically open	1658	12.16 (2.46)	-0.58*** (0.13)	-0.14 (0.19)	-0.16 (0.22)
Applied for loans in past 12 months	1658	0.17 (0.38)	-0.01 (0.03)	0.05 (0.03)	-0.01 (0.04)
Total value of outstanding loans (USD)	1596	350.34 (1174.33)	-11.06 (76.81)	98.83 (90.68)	-183.34* (110.30)

This table shows the difference-in-difference results from the main equation. Each row presents results from one regression with a different socio-economic variable as the outcome. All variables measuring values are in USD. Profit is measured by directly asking the respondent, rather than by subtracting costs from revenues. Total reported costs are the sum of costs for specific items/activities and are not comprehensive. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is the business owner or a manager, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. The effects on business costs furthermore do not survive False Discovery Rate (FDR) adjustment for multiple hypothesis testing (Table C8). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B7: Impact of transformer injection on household outcomes

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post $\times$ Treat (SE)
Last month HH income (USD)	1358	332.33 (473.05)	-34.24 (36.83)	16.93 (41.60)	-75.43 (49.79)
Monthly rent (USD)	490	30.78 (25.14)	-3.86*** (1.29)	1.28 (3.46)	-0.52 (1.93)
Share of HH adults (18+) with paid jobs in last 7 days	1484	0.65 (0.36)	-0.02 (0.03)	0.03 (0.03)	-0.04 (0.04)
Household use of dirty cooking fuel (past 3 months)	1492	0.65 (0.48)	0.08*** (0.02)	-0.03 (0.03)	0.04 (0.04)
Last 2 weeks spending on health (USD)	1408	13.22 (34.91)	4.70 (2.88)	-1.37 (2.30)	3.83 (3.91)
Household qualitative assessments index	1492	1.38 (2.26)	-1.38*** (0.16)	0.02 (0.20)	0.03 (0.23)
Perceived safety in area (1-5)	1490	3.51 (0.97)	0.02 (0.07)	-0.17** (0.09)	0.05 (0.12)
Belief that Dumsor is back (1-5)	1486	2.99 (1.27)	1.32*** (0.10)	-0.00 (0.12)	0.07 (0.13)
Expected reliability one year from today (1-3)	1032	2.33 (0.79)	0.32*** (0.07)	0.02 (0.08)	0.02 (0.10)
Loss of perishable food due to reliability (0-2)	1486	0.33 (0.54)	-0.31*** (0.03)	-0.02 (0.05)	0.04 (0.05)
Loss of perishable medicine due to reliability (0-2)	1486	0.04 (0.20)	-0.03*** (0.01)	0.00 (0.02)	-0.00 (0.02)
Household health challenges due to reliability issues	1484	0.00 (1.00)	-0.00 (0.07)	0.07 (0.08)	-0.02 (0.12)
Household study light quality index	1492	0.00 (1.00)	-0.04 (0.06)	-0.06 (0.06)	0.03 (0.07)
Hours per day lightbulbs used for reading or studying	402	0.90 (0.24)	0.08*** (0.03)	-0.00 (0.03)	-0.00 (0.04)
Share of hours per day reading or studying with lightbulbs	1492	0.13 (0.59)	-0.09*** (0.03)	-0.04 (0.04)	0.03 (0.04)

This table shows the difference-in-difference results from the main equation. Each row presents results from one regression with a different socio-economic variable as the outcome. Total household monthly income reflects the sum of incomes from any source for all household members of age 16 and above. Monthly rent is missing for individuals who do not rent or occupy the premises rent free. Dirty cooking fuel includes wood, charcoal, and animal waste, but not gas, electricity, or kerosene. All variables measuring values are in USD. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, the count of all household members and of household adults, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B8: Impacts of transformer injection intervention on average voltage by baseline voltage quality

	(1)	(2)	(3)	(4)
	All sites	Below median baseline voltage	Above median baseline voltage	All sites
Post construction	5.94*** (1.74)	10.02*** (2.99)	1.85 (1.32)	1.98 (1.31)
During construction	0.76 (1.09)	2.04 (2.15)	-0.01 (1.07)	0.32 (1.06)
Treat X Post	5.48** (2.48)	8.70** (4.09)	2.20 (2.00)	2.22 (2.00)
Treat X During	2.38 (1.60)	5.65* (2.84)	-0.99 (1.48)	-0.99 (1.47)
Post construction=1 $\times$ PBQ=1				7.92** (3.23)
During construction=1 $\times$ PBQ=1				1.44 (2.29)
Treat X Post=1 $\times$ PBQ=1				6.46 (4.54)
Treat X During=1 $\times$ PBQ=1				6.61** (3.19)
Observations	9866078	5258541	4607537	9866078
Pre-construction control mean	219.18	210.02	227.71	227.71
Hour of day FE	Y	Y	Y	Y
Week of year FE	Y	Y	Y	Y
Site FE	Y	Y	Y	Y

This table shows the difference-in-differences estimates by baseline voltage quality, measured as the mean share of the time in each site that voltage was within 10% of nominal. Each column presents results from one regression with average voltage as the outcome variable, but for different samples. Subsetting by baseline voltage is done separately for treatment and control sites, so the samples always include an equal number of each. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B9: Impacts of transformer injection intervention on share of bad voltage hours by baseline voltage quality

	(1)	(2)	(3)	(4)
	All sites	Below median baseline voltage	Above median baseline voltage	All sites
Post construction	-0.08*** (0.02)	-0.13*** (0.04)	-0.02 (0.01)	-0.02* (0.01)
During construction	-0.01 (0.02)	-0.02 (0.03)	-0.00 (0.02)	-0.01 (0.02)
Treat X Post	-0.10*** (0.03)	-0.19*** (0.05)	-0.00 (0.02)	-0.00 (0.02)
Treat X During	-0.06** (0.03)	-0.15*** (0.05)	0.03 (0.03)	0.03 (0.03)
Post construction=1 $\times$ PBQ=1				-0.10** (0.04)
During construction=1 $\times$ PBQ=1				0.00 (0.03)
Treat X Post=1 $\times$ PBQ=1				-0.19*** (0.06)
Treat X During=1 $\times$ PBQ=1				-0.18*** (0.05)
Observations	9866078	5258541	4607537	9866078
Pre-construction control mean	0.21	0.36	0.07	0.07
Hour of day FE	Y	Y	Y	Y
Week of year FE	Y	Y	Y	Y
Site FE	Y	Y	Y	Y

This table shows the difference-in-differences estimates by baseline voltage quality, measured as the mean share of the time in each site that voltage was within 10% of nominal. Each column presents results from one regression with the share of time voltage was at least 10% below nominal as the outcome variable, but for different samples. Subsetting by baseline voltage is done separately for treatment and control sites, so the samples always include an equal number of each. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B10: Heterogeneous impacts of transformer injection intervention on primary outcomes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Good voltage (device)	Hours reported bad voltage (above median)	Business electricity importance (high)	WTP for reliability (above median)	Defensive investments (high)	Appliances owned (above median)	Fridge count (≥ 1)	Worst
Voltage damage and protection index	0.05 (0.12)	-0.03 (0.12)	-0.04 (0.12)	-0.07 (0.10)	-0.05 (0.13)	-0.01 (0.10)	0.07 (0.10)	0.02 (0.15)
Any voltage-related damage, last 12 months (=1)	0.03 (0.07)	-0.02 (0.07)	-0.01 (0.08)	-0.00 (0.06)	-0.02 (0.07)	-0.01 (0.06)	0.02 (0.06)	0.04 (0.09)
Any reliability protective device owned (=1)	-0.00 (0.02)	-0.00 (0.03)	-0.01 (0.04)	-0.05* (0.03)	-0.01 (0.04)	0.00 (0.02)	0.02 (0.02)	-0.03 (0.03)
Amt. spent on burnt/broken apps in past year (USD)	-2.39 (3.70)	2.99 (3.73)	-1.21 (3.73)	-4.10 (3.80)	0.16 (5.00)	-1.52 (3.82)	1.19 (3.51)	0.49 (5.14)
Value of protective devices owned (USD)	-5.20* (3.12)	2.74 (3.86)	0.84 (3.41)	-2.44 (3.43)	4.87 (12.02)	2.82 (3.64)	4.48 (3.04)	-2.03 (2.36)
Reported hours of bad voltage in past month	-14.76 (14.25)	-9.49 (13.23)	-2.16 (13.82)	4.17 (11.25)	-6.62 (12.79)	-12.03 (9.30)	-0.59 (10.15)	-7.54 (16.13)
Reported total outage hours in past month	-0.30 (5.00)	3.02 (3.96)	3.24 (4.83)	-1.71 (3.32)	-5.32 (3.80)	-6.59* (3.38)	-3.15 (3.36)	-1.51 (5.42)
WTP for perfect reliability and quality (USD)	1.11** (0.53)	1.06** (0.52)	0.59 (0.74)	0.83** (0.42)	-0.11 (0.55)	0.54 (0.52)	0.49 (0.41)	2.13*** (0.76)
WTP for perfect voltage and half outage hours (USD)	1.03** (0.41)	0.74** (0.37)	0.13 (0.46)	0.35 (0.33)	-0.14 (0.42)	0.17 (0.36)	0.14 (0.32)	1.20** (0.56)
WTP for no outages and half bad voltage hours (USD)	2.61*** (0.86)	1.68* (0.87)	2.21 (1.46)	0.90 (0.83)	-0.45 (1.18)	1.00 (0.92)	0.12 (0.66)	1.72 (1.36)
Total no. of appliances owned	-0.13 (0.24)	-0.02 (0.26)	-0.13 (0.40)	-0.36 (0.25)	-0.07 (0.33)	0.18 (0.25)	0.07 (0.22)	-0.07 (0.28)
Any alt. energy source used in last month (=1)	-0.01 (0.02)	-0.00 (0.02)	0.06** (0.03)	-0.03 (0.02)	-0.02 (0.02)	0.00 (0.02)	-0.01 (0.02)	0.03 (0.04)
Last month electricity spending (USD)	-0.58 (1.57)	0.89 (1.31)	-2.22 (1.84)	-0.52 (1.62)	-1.30 (1.87)	-0.79 (1.49)	-1.87 (1.45)	0.57 (2.04)
Last month business profit (USD)	46.86 (29.48)	6.49 (26.10)	24.00 (29.91)	5.95 (28.23)	-22.31 (29.62)	-27.12 (30.61)	14.45 (24.91)	40.57 (56.67)
Last month business revenue (USD)	237.17** (114.63)	40.40 (101.19)	231.01 (156.52)	-38.59 (112.21)	-172.62 (140.55)	6.62 (112.29)	66.93 (114.66)	128.46 (196.55)
Last month business costs (USD)	66.52 (96.66)	39.74 (82.34)	168.46 (121.84)	-51.50 (94.85)	-109.27 (105.08)	-29.68 (99.49)	-44.15 (91.97)	-30.60 (144.59)
Last month HH income (USD)	42.59 (94.35)	29.57 (87.67)	0.00 (.)	-61.55 (86.75)	14.91 (101.05)	35.72 (80.13)	40.69 (82.82)	150.02 (115.63)

Difference-in-difference estimates of triple interactions with specific baseline customer characteristics. Each cell is one regression. The coefficients correspond to α_1 from the following equation: $Y_{it} = \alpha_0 + \alpha_1 \text{Group} * \text{Treat} * \text{Post}_{it} + \alpha_2 \text{Group} * \text{Treat}_{it} + \alpha_3 \text{Group} * \text{Post}_{it} + \alpha_4 \text{Post} * \text{Treat}_{it} + \alpha_5 \text{Post}_t + \alpha_6 \text{Treat}_i + \alpha_7 \text{Group}_i + u_{it}$. Good voltage equals 1 for those that are below the median per device measurements. Hours of reported bad voltage equals 1 for those reporting levels of bad voltage above the median at baseline. “High electricity importance =1” if the owner reported that electricity is “very important” or “extremely important”. “High defensive investments” means having at least 1 reliability protective device. Appliances owned equals to 1 if a respondent has more appliances than the median. Fridge count equals 1 if the respondent has at least one fridge at baseline. “Worst” equals 1 if at baseline, the respondent has below median voltage and owns appliances sensitive to voltage but does not own protective devices. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

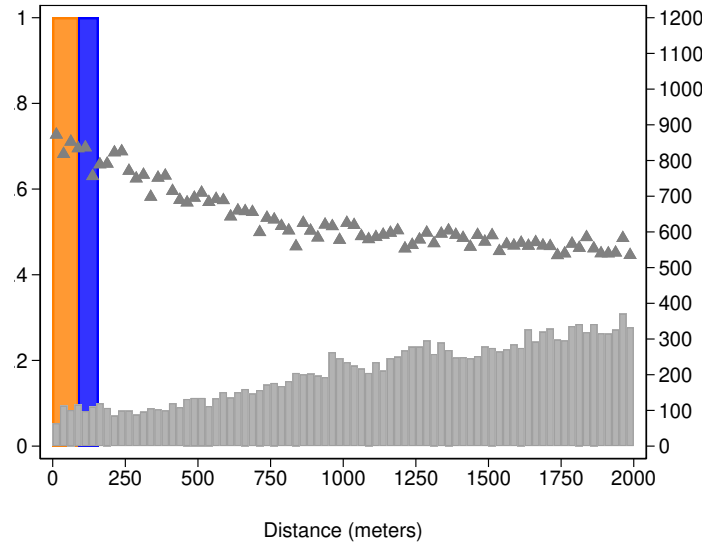
Table B11: Heterogeneous impacts of transformer injection intervention by level of outages

	(1)	(2)	(3)	(4)
	Above median pre-LB monthly outage hours (device)	Above median baseline last month outage hours (resp.)	Above median post-LB monthly outage hours (device)	Above median endline last month outage hours (resp.)
Voltage damage and protection index	-0.19 (0.12)	0.04 (0.10)	0.01 (0.12)	-0.07 (0.12)
Any voltage-related damage, last 12 months (=1)	-0.12 (0.07)	0.03 (0.06)	0.01 (0.07)	0.00 (0.07)
Any reliability protective device owned (=1)	-0.00 (0.03)	-0.00 (0.03)	-0.01 (0.02)	-0.05* (0.03)
Amt. spent on burnt/broken apps in past year (USD)	-0.72 (3.75)	-0.24 (3.74)	-0.00 (3.50)	3.65 (3.72)
Value of protective devices owned (USD)	-4.94 (3.42)	-5.07 (3.29)	-6.37* (3.26)	-1.45 (3.17)
Reported hours of bad voltage in past month	1.32 (13.70)	0.50 (11.15)	-2.91 (13.34)	2.26 (8.61)
Reported total outage hours in past month	0.58 (4.78)	4.88 (3.28)	-5.56 (4.74)	-5.75 (3.78)
WTP for perfect reliability and quality (USD)	-0.81 (0.52)	-1.04** (0.48)	0.10 (0.54)	-0.16 (0.46)
WTP for perfect voltage and half outage hours (USD)	-0.13 (0.39)	-0.75** (0.35)	0.15 (0.41)	0.23 (0.32)
WTP for no outages and half bad voltage hours (USD)	-0.73 (0.78)	-0.15 (0.80)	0.89 (0.90)	0.53 (0.82)
Total no. of appliances owned	0.22 (0.28)	-0.14 (0.25)	0.22 (0.24)	0.27 (0.23)
Plans to purchase >=1 new appliance in next year	0.07 (0.07)	-0.00 (0.07)	-0.04 (0.07)	-0.07 (0.07)
Any alt. energy source used in last month (=1)	0.01 (0.02)	0.02 (0.02)	-0.01 (0.02)	-0.00 (0.02)
Last month electricity spending (USD)	-0.08 (1.61)	2.83* (1.61)	0.10 (1.56)	0.73 (1.26)
Last month business profit (USD)	-28.83 (29.59)	1.24 (26.47)	-65.16** (26.78)	-41.74* (24.00)
Last month business revenue (USD)	-36.17 (118.44)	50.83 (107.38)	-130.41 (110.96)	-118.84 (103.45)
Last month business costs (USD)	-63.33 (100.62)	23.01 (79.95)	-11.49 (100.15)	-55.01 (88.88)
Last month HH income (USD)	100.80 (104.22)	-151.39* (86.29)	-19.78 (89.52)	-122.37 (87.22)

Difference-in-difference estimates of triple interactions with specific baseline customer characteristics. Each cell is one regression. The coefficients correspond to α_1 from the following equation: $Y_{it} = \alpha_0 + \alpha_1 \text{Group} * \text{Treat} * \text{Post}_{it} + \alpha_2 \text{Group} * \text{Treat}_{it} + \alpha_3 \text{Group} * \text{Post}_{it} + \alpha_4 \text{Post} * \text{Treat}_{it} + \alpha_5 \text{Post}_t + \alpha_6 \text{Treat}_i + \alpha_7 \text{Group}_i + u_{it}$. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Appendix C: Robustness checks

Figure C1: Power quality is highly correlated within short distances



There are 178,241 pairs of devices with at least one month of overlap. For each pair we calculate distance and the correlation in hourly voltage data. The graph plots the average correlation coefficient and number of pairs in 25m bins. 50% of respondents are $\leq 88\text{m}$ of a device (in orange/darker shaded band); 30% of respondents are between 88–157m (in blue/lighter shaded band).

Table C1: Balance between panel and attrited respondents

	Matched N	Mean	Attrited N	Difference	<i>p</i> -val
<i>Respondent and Location</i>					
Age (years)	1575	39.23	426	2.73	0.000
Respondent is male	1575	0.35	426	0.02	0.547
Completed secondary education	1575	0.50	426	-0.05	0.050
Owns premises	1575	0.38	426	0.12	0.000
<i>Appliances</i>					
Any television (TV) at location	1575	0.71	426	-0.02	0.419
Any fridge at location	1575	0.62	426	0.01	0.835
Count of mobile phones	1575	2.17	426	0.05	0.605
Any electricity protective devices	1575	0.25	426	0.01	0.571
Count of reliability protective devices	1575	0.37	426	0.01	0.785
<i>Electricity and Energy</i>					
Pays someone else for electricity	1575	0.09	426	-0.08	0.000
Count of meter users	1566	1.84	424	-0.44	0.001
Last month electricity spending (USD)	1559	16.68	413	0.46	0.615
Has generator	1575	0.04	426	0.02	0.070
Count of alternative fuels used in past 3 months	1575	0.91	426	-0.07	0.107
Last month spending on alternative fuels (USD)	1575	6.24	426	-0.30	0.497
Reported total outage hours in past month	1575	39.88	426	3.44	0.151
Reported hours of bad voltage in past month	1566	47.49	422	4.29	0.408
Any burnt/broken appliances in past 12 months	1575	0.27	426	0.03	0.141
Amt. spent on burnt/broken apps in past year (USD)	1562	9.30	426	4.10	0.002
<i>Household Characteristics</i>					
Adults	746	2.39	251	0.27	0.002
Children (<18)	746	1.21	251	0.24	0.014
Last month HH income (USD)	714	352.97	234	-34.53	0.502
Share of HH adults with paid jobs in last week	746	0.66	251	-0.04	0.123
<i>Business Characteristics</i>					
Number of workers	829	1.97	175	-0.11	0.579
Last month business revenue (USD)	723	398.22	147	-65.65	0.292
Last month business costs (USD)	829	272.95	175	-29.26	0.449
Last month business profit (USD)	646	98.15	131	-31.02	0.063
Usual business open hours	829	12.07	175	0.20	0.397
Any non-electric business machines at location	829	0.09	175	0.00	0.992
Business engaged in retail activities (=1)	829	0.44	175	-0.07	0.085
Business engaged in manufacturing activities (=1)	829	0.20	175	0.11	0.000
Business engaged in other service activities (=1)	829	0.36	175	-0.03	0.409
Business activity likely using electricity (=1)	829	0.22	175	0.09	0.004

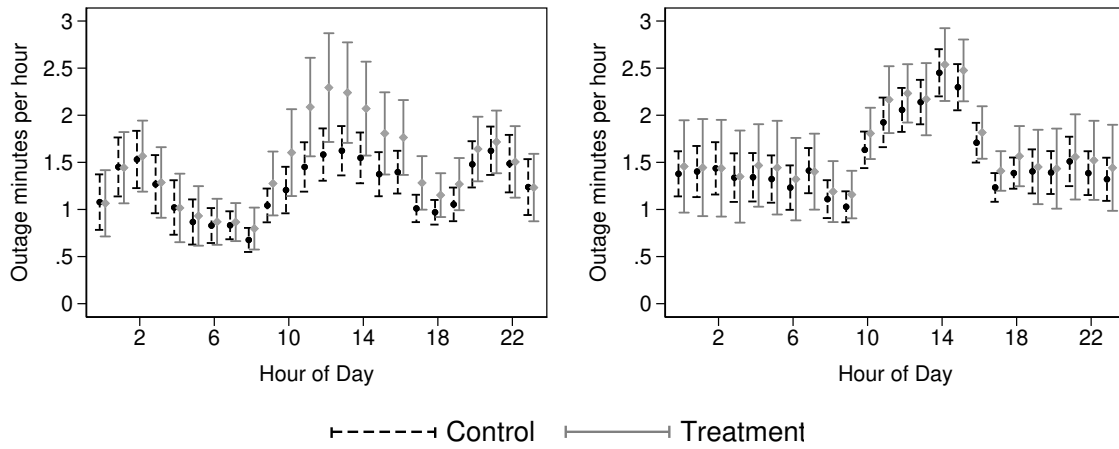
This table shows means in the baseline period for survey respondents, pooling businesses and households, and tests for significance of the differences in means by whether the respondent was also surveyed at the endline. The *p*-value for the joint F-test for household baseline characteristics is <0.001. The *p*-value for the joint F-test for business baseline characteristics is 0.024. * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01

Table C2: Correlation between attrited respondents' characteristics and treatment status

	Mean	LB Treat	N
Age (years)	37.11 [12.40]	-1.20 (1.09)	426
Respondent is male	0.37 [0.48]	-0.07 (0.05)	426
Completed secondary education	0.54 [0.50]	0.04 (0.05)	426
Owns premises	0.31 [0.47]	-0.12*** (0.04)	426
Any television (TV) at location	0.69 [0.47]	0.08* (0.04)	426
Any fridge at location	0.60 [0.49]	0.02 (0.05)	426
Count of mobile phones	2.22 [2.00]	-0.20 (0.18)	426
Any electricity protective devices	0.26 [0.44]	-0.03 (0.04)	426
Count of reliability protective devices	0.37 [0.70]	-0.02 (0.07)	426
Pays someone else for electricity	0.17 [0.38]	-0.00 (0.04)	426
Count of meter users	2.35 [2.94]	-0.13 (0.24)	424
Last month electricity spending (USD)	17.12 [16.58]	-1.76 (1.64)	413
Has generator	0.02 [0.14]	0.01 (0.02)	426
Count of alternative fuels used in past 3 months	0.96 [0.82]	0.05 (0.08)	426
Last month spending on alternative fuels (USD)	6.21 [7.70]	0.63 (0.78)	426
Reported total outage hours in past month	35.39 [40.27]	2.05 (4.10)	426
Reported hours of bad voltage in past month	42.73 [102.40]	0.90 (9.27)	422
Any burnt/broken appliances from voltage fluctuations in past 12 months	0.24 [0.43]	-0.01 (0.04)	426
Amt. spent on burnt/broken apps in past year (USD)	4.59 [19.44]	1.19 (2.02)	426

This table shows the correlation between the attrited respondents' characteristics and treatment status. The sample is restricted to respondents who do not participate in the endline survey. We regress each respondent characteristic at baseline on a dummy variable equals one if the respondent was in a treatment site at baseline. Each row represents an outcome. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure C2: Impacts of transformer injection on outages by time of day
A) Before construction B) After construction



The figure shows mean minutes of power outages by hour of day separately for treatment and control sites. 95% confidence intervals around treatment means are clustered at the site level. Panel A shows means for the year prior to the start of the transformer construction period, and Panel B shows means for the year after the end of the construction period. [Figure 4](#) shows impacts on average voltage.

Table C3: Impact of transformer injection intervention on hourly average voltage, robustness to implementation issues

	(1)	(2)	(3)	(4)	(5)
	All sites	Commissioned sites (SMEC)	New tx confirmed sites	All sites, IV new tx with treat	All sites
During construction	0.76 (1.09)	0.79 (1.09)	0.65 (1.18)	0.42 (1.22)	0.79 (1.55)
Treat X During	2.38 (1.60)	2.55 (1.63)	3.61** (1.70)		1.95 (1.93)
Post construction	5.94*** (1.74)	5.95*** (1.74)	5.26*** (1.85)	5.26*** (1.95)	5.95*** (1.82)
Treat X Post	5.48** (2.48)	6.06** (2.49)	8.41*** (2.48)		5.46** (2.59)
New transformer X Post				7.13** (3.13)	
New transformer X During				3.24 (2.05)	
Observations	9866078	9723260	8815828	9866078	9866078
Pre-constr. ctl. mean	219.18	219.18	220.24	219.18	217.95
Hour of day FE	Y	Y	Y	Y	Y
Week of year FE	Y	Y	Y	Y	Y
Site FE	Y	Y	Y	Y	Y
Revised constr. period	N	N	N	N	Y

This table shows the difference-in-difference results for the impact of the transformer injection treatment on hourly average voltage levels measured by GridWatch devices in each site. Each column is one regression. Column 2 drops two sites where the construction manager SMEC indicated the new transformer was not commissioned successfully. Column 3 drops sites where our own construction monitoring activities indicated no new transformer was built in a treatment site or a new transformer was built in a control site. Column 4 instruments for observing a new transformer during the construction monitoring visits with site treatment assignment. Column 5 defines the construction period as July 1, 2020-December 31, 2020 instead of October 1, 2020-March 31, 2021, based on reported dates of transformer construction activity from the construction manager, SMEC. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. [Table 5](#) presents the main specification.

Table C4: Impact of transformer injection intervention on hourly outage minutes, robustness to implementation issues

	(1)	(2)	(3)	(4)	(5)
	All sites	Commissioned sites (SMEC)	New tx confirmed sites	All sites, IV new tx with treat	All sites
During construction	0.21*** (0.07)	0.21*** (0.07)	0.16** (0.08)	0.22*** (0.08)	0.00 (0.08)
Treat X During	-0.06 (0.12)	-0.07 (0.12)	-0.04 (0.13)		-0.11 (0.12)
Post construction	-0.08 (0.08)	-0.08 (0.08)	-0.12 (0.08)	-0.05 (0.09)	-0.04 (0.08)
Treat X Post	-0.21 (0.13)	-0.23* (0.13)	-0.22 (0.14)		-0.23* (0.14)
New Trafo X Post				-0.28 (0.17)	
New Trafo X During				-0.09 (0.15)	
Observations	10033086	9888612	8962703	10033086	10033086
Pre-constr. ctl. mean	1.39	1.39	1.41	1.39	1.48
Hour of day FE	Y	Y	Y	Y	Y
Week of year FE	Y	Y	Y	Y	Y
Site FE	Y	Y	Y	Y	Y
Revised constr. period	N	N	N	N	Y

This table shows the difference-in-difference results for the impact of the transformer injection treatment on hourly power outage minutes measured by GridWatch devices in each site. Each column is one regression. Column 2 drops two sites where the construction manager SMEC indicated the new transformer was not commissioned successfully. Column 3 drops sites where our own construction monitoring activities indicated no new transformer was built in a treatment site or a new transformer was built in a control site. Column 4 instruments for observing a new transformer during the construction monitoring visits with site treatment assignment. Column 5 defines the construction period as July 1, 2020-December 31, 2020 instead of October 1, 2020-March 31, 2021, based on reported dates of transformer construction activity from the construction manager, SMEC. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. [Table 5](#) presents the main specification.

Table C5: Impacts of transformer injection intervention on additional voltage quality measures, hourly data

	N	Control Mean (SD)	Post (SE)	Treat $\times$ Post (SE)
Mean voltage during hour	9866078	219.18 (22.39)	5.94*** (1.74)	5.48** (2.48)
Absolute deviation from nominal	9866078	16.51 (18.60)	-3.74*** (1.14)	-0.83 (1.70)
Any voltage >20% below nominal	10033086	0.08 (0.26)	-0.03** (0.01)	-0.04** (0.02)
Minutes voltage >20% above nominal	10033086	0.00 (0.31)	-0.00 (0.00)	-0.00 (0.01)
Minutes voltage 10-20% below nominal	10033086	9.60 (19.83)	-2.96*** (0.93)	-4.21*** (1.41)
Minutes voltage >20% below nominal	10033086	2.76 (11.59)	-1.54*** (0.57)	-1.40* (0.85)
Minutes with no power (outage)	10033086	1.39 (8.59)	-0.08 (0.08)	-0.21 (0.13)

This table shows the difference-in-differences effects of the transformer injection intervention on measures of voltage quality using hourly data at the GridWatch device level. Each row is one regression. The minutes variables indicate the number of minutes in each hourly observation that the electricity had a certain status. “Any voltage >20% below nominal” is a dummy variable for whether voltage fell below this threshold at any point during an hourly observation. All regressions include hour of day, week of year, and site fixed effects. Standard errors are clustered at the site level.

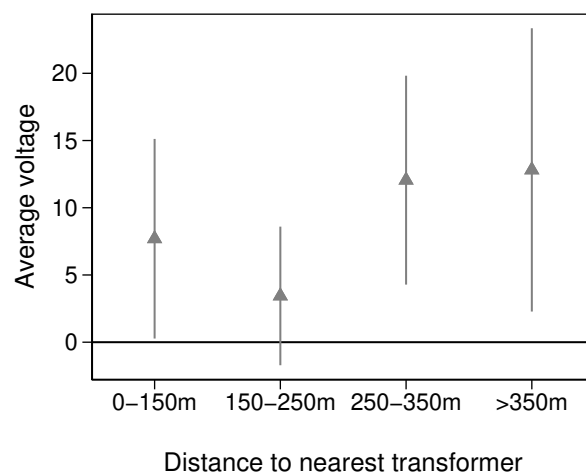
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. [Table 5](#) presents the main specification.

Table C6: Impacts of transformer injection intervention on voltage quality, monthly data

	N	Control Mean (SD)	Post (SE)	Treat $\times$ Post (SE)
Hours with voltage >20% above nominal	14213	0.03 (0.47)	-0.01 (0.01)	-0.05 (0.06)
Hours with voltage 10-20% below nominal	14213	114.27 (152.16)	-33.31*** (11.24)	-52.00*** (17.15)
Hours with voltage >20% below nominal	14213	32.81 (99.40)	-18.18** (7.11)	-16.28 (10.58)
Hours with no power (outages)	14213	13.52 (13.60)	-0.45 (0.73)	-2.12 (1.40)
Number of spells with voltage <207	14213	251.92 (300.46)	-68.51*** (20.10)	-67.83** (31.18)
Number of spells with min voltage >200	14213	212.82 (255.67)	-60.65*** (16.55)	-50.46* (25.88)
Number of spells with min voltage btwn 184-200	14213	29.88 (46.22)	-6.15* (3.67)	-12.80** (5.19)
Number of spells with min voltage <184	14213	9.22 (14.45)	-1.70* (1.02)	-4.57*** (1.67)
Total duration of spells with voltage <207	14213	125.07 (181.45)	-43.46*** (14.11)	-54.68*** (20.80)
Total duration of spells with min voltage >200	14213	15.88 (18.97)	-3.67*** (1.25)	-3.98** (1.89)
Total duration of spells with min voltage btwn 184-200	14213	32.51 (45.81)	-7.76** (3.07)	-8.71** (4.23)
Total duration of spells with min voltage <184	14213	76.69 (154.91)	-32.03*** (11.31)	-41.99** (16.82)
Share of low-voltage time in spells with min voltage <184	11975	0.37 (0.38)	0.03 (0.02)	0.02 (0.04)
Mean spell length (hours)	11975	0.74 (2.19)	-0.29* (0.17)	-0.08 (0.23)
Median spell length (hours)	11975	0.12 (0.58)	-0.05 (0.04)	0.06 (0.05)
Maximum spell length (hours)	11975	11.98 (22.38)	-3.42** (1.72)	-5.90** (2.65)
Mean of mean voltage during a spell	11975	189.84 (35.33)	-8.33*** (2.50)	-15.76*** (4.09)
Median of mean voltage during a spell	11975	191.94 (36.10)	-7.95*** (2.54)	-15.70*** (4.10)
Mean of minimum voltage during a spell	11975	187.41 (36.69)	-8.54*** (2.63)	-15.32*** (4.30)

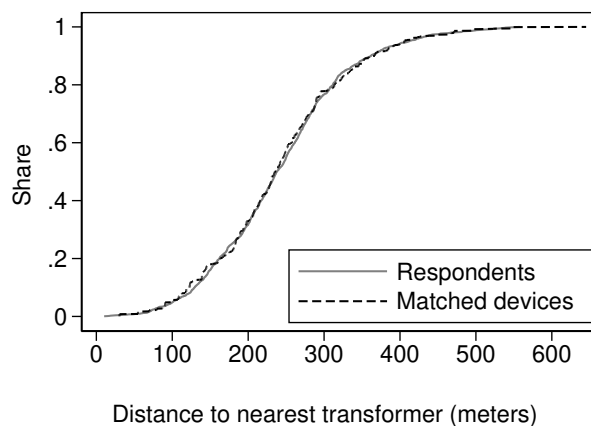
This table shows the difference-in-differences effects of the transformer injection intervention on measures of voltage quality using monthly data at the GridWatch device level. Each row is one regression. Outcomes in all but the first row are measured based on identifying individual low-voltage “spells” during which voltage fell below 207V (10% below nominal) in any 2-minute interval. Individual spells with different characteristics are then aggregated to the month-device level. Months where there were no low-voltage spells for particular devices are assigned a 0 for outcomes that are not conditional on experiencing at least one such spell. Number of spells refers to the number of individual low-voltage spells in a device-month. Total hours of spells take the sum of the duration of individual spells in a device-month. Mean, median, and maximum spell length are statistics calculated over all individual spells in a device-month. Mean and median of mean spell voltage are statistics calculated over the mean voltage level within a spell for all individual spells in a device-month. Mean minimum voltage is calculated similarly. These statistics are conditional on any low-voltage spell being observed in a device-month. All regressions include month and site fixed effects. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. [Table 5](#) presents the same using hourly data.

Figure C3: Impact of transformer injection intervention by distance to the nearest transformer at baseline



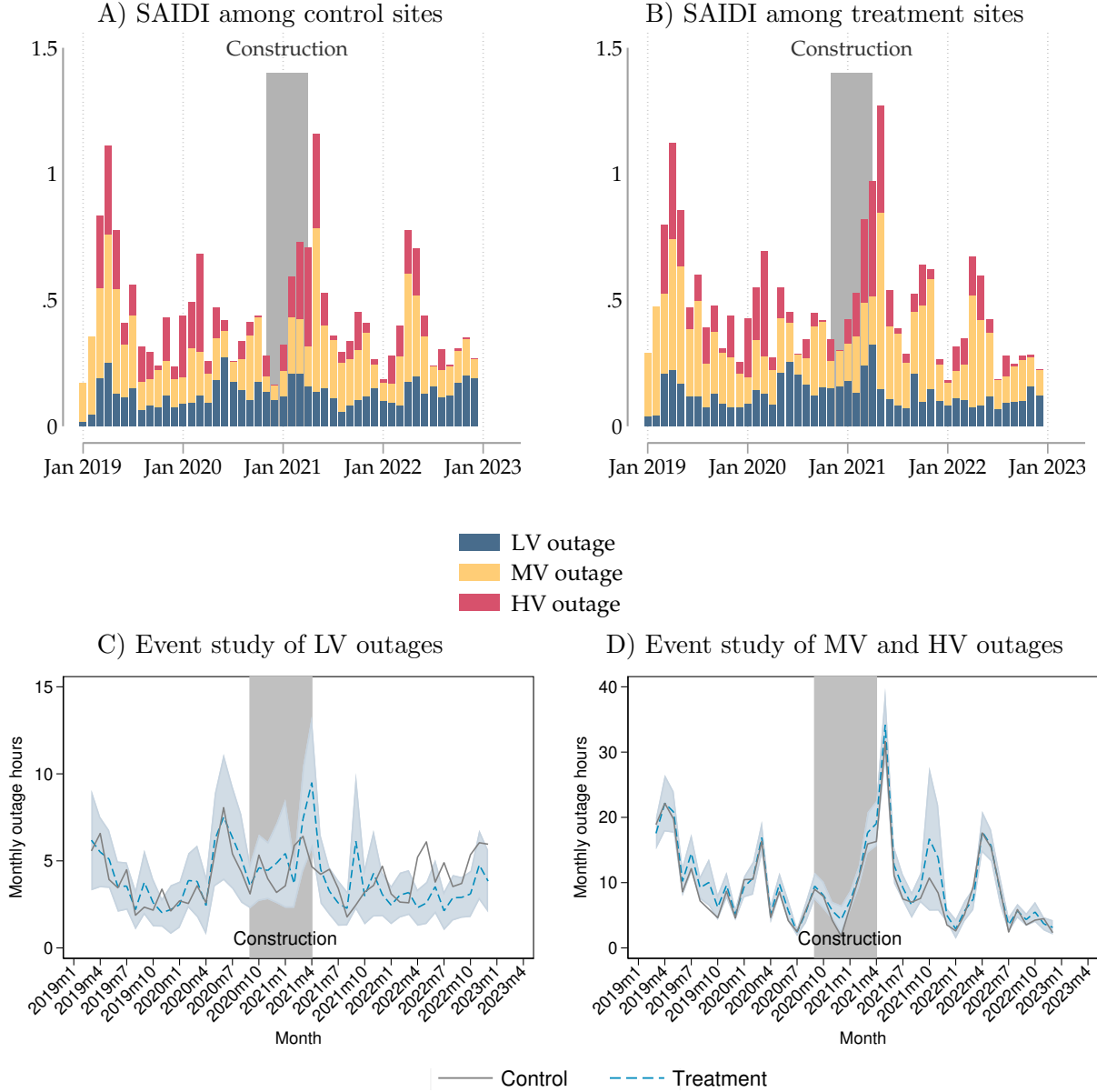
The figure shows coefficients for the differential impacts on average voltage of baseline distance of the GridWatch device from the nearest transformer in treatment sites relative to control sites in the post-construction period. A joint test cannot reject equality of the four treatment-effect estimates across baseline-distance categories ($F=2.03$, $p=0.113$). [Table 5](#) presents the pooled specification using hourly data.

Figure C4: Distribution of distances between respondents and devices to the nearest transformer



Distance to the nearest transformer is measured during the baseline period before new transformers were added in treatment sites. The sample of devices includes only those that are matched to respondents as being the closest device.

Figure C5: Impacts on low-voltage, medium-voltage, and high-voltage power outages



Notes: Panels A and B plot the System Average Interruption Duration Index (SAIDI) for control sites and treatment sites by whether the outage originates at the low-voltage, medium-voltage, or high-voltage part of the electricity network. Panels C and D plot event study plots for the impact of the intervention on low-voltage outages and on medium- and high-voltage outages, respectively.

Table C7: Impact of transformer injection intervention on primary outcomes for businesses

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post $\times$ Treat (SE)
Voltage damage and protection index	1658	-0.15 (0.93)	-0.07 (0.06)	0.13 (0.08)	-0.08 (0.08)
Any voltage-related damage, last 12 months (=1)	1658	0.22 (0.41)	-0.02 (0.03)	0.04 (0.04)	-0.05 (0.04)
Any reliability protective device owned (=1)	1658	0.19 (0.39)	-0.03*** (0.01)	0.04 (0.03)	-0.01 (0.02)
Amt. spent on burnt/broken apps in past year (USD)	1628	7.18 (27.84)	-3.80** (1.62)	-0.28 (2.05)	1.66 (2.21)
Value of protective devices owned (USD)	1426	5.14 (27.95)	-3.10** (1.53)	-0.55 (2.25)	-0.23 (2.22)
Reported hours of bad voltage in past month	1636	40.61 (82.68)	-39.84*** (4.38)	8.28 (7.83)	-10.11 (8.38)
Reported total outage hours in past month	1619	31.99 (31.34)	-29.10*** (2.28)	-0.40 (3.01)	1.31 (3.07)
WTP for perfect reliability and quality (USD)	1658	3.10 (4.35)	-1.07*** (0.25)	-0.26 (0.30)	0.33 (0.35)
WTP for perfect voltage and half outage hours (USD)	1658	1.58 (2.78)	-0.26 (0.19)	-0.27 (0.20)	0.34 (0.25)
WTP for no outages and half bad voltage hours (USD)	440	1.88 (3.46)	0.34 (0.41)	-0.26 (0.43)	0.88 (0.59)
Total no. of appliances owned	1658	7.00 (5.54)	-0.16 (0.13)	-0.06 (0.38)	0.22 (0.18)
Any alt. energy source used in last month (=1)	1658	0.06 (0.24)	-0.01 (0.01)	-0.00 (0.02)	0.00 (0.01)
Last month electricity spending (USD)	1594	17.63 (16.99)	-3.92*** (0.73)	-1.74 (1.22)	0.21 (0.96)

This table shows the difference-in-difference results from [Table B5](#) for businesses only. Each row is one regression. All variables measuring values are in USD. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. [Table D3](#) shows additional business outcomes.

Table C8: Impact of transformer injection intervention on primary outcomes

	N	Control Mean (SD)	Post (SE)	FDR q-value	Treat (SE)	FDR q-value	Post $\times$ Treat (SE)	FDR q-value
Voltage damage and protection index	3150	0.00 (1.00)	-0.11** (0.05)	0.044	0.09 (0.06)	0.718	-0.10* (0.06)	0.350
Any voltage-related damage, last 12 months (=1)	3150	0.25 (0.43)	-0.05* (0.03)	0.120	0.04 (0.03)	0.718	-0.05 (0.04)	0.432
Any reliability protective device owned (=1)	3150	0.25 (0.44)	-0.02** (0.01)	0.032	0.02 (0.02)	0.775	-0.02 (0.01)	0.350
Amt. spent on burnt/broken apps in past year (USD)	3080	9.28 (33.80)	-5.77*** (1.37)	0.001	0.05 (1.77)	0.988	1.23 (1.91)	0.668
Value of protective devices owned (USD)	2668	5.49 (23.16)	-2.32** (0.92)	0.032	0.36 (1.42)	0.941	0.32 (1.63)	0.895
Reported hours of bad voltage in past month	3110	43.18 (87.51)	-42.48*** (4.72)	0.001	7.60 (7.41)	0.718	-9.09 (7.70)	0.432
Reported total outage hours in past month	3081	32.15 (31.08)	-29.32*** (2.05)	0.001	1.69 (2.60)	0.775	-1.18 (2.74)	0.802
WTP for perfect reliability and quality (USD)	3150	3.29 (4.41)	-1.41*** (0.21)	0.001	-0.34 (0.25)	0.718	0.44 (0.28)	0.350
WTP for perfect voltage and half outage hours (USD)	3150	1.58 (2.71)	-0.37** (0.16)	0.044	-0.17 (0.17)	0.718	0.21 (0.21)	0.529
WTP for no outages and half bad voltage hours (USD)	850	1.95 (3.34)	-0.07 (0.30)	0.837	-0.37 (0.32)	0.718	0.99** (0.44)	0.350
Total no. of appliances owned	3150	8.55 (5.99)	-0.02 (0.09)	0.837	-0.07 (0.35)	0.941	0.09 (0.13)	0.668
Plans to purchase ≥ 1 new appliance in next year	3150	0.58 (0.49)	-0.05* (0.03)	0.120	-0.02 (0.03)	0.775	0.04 (0.04)	0.432
Any alt. energy source used in last month (=1)	3150	0.05 (0.22)	-0.01 (0.01)	0.409	0.01 (0.01)	0.881	-0.00 (0.01)	0.933
Last month electricity spending (USD)	3050	17.72 (16.95)	-3.92*** (0.60)	0.001	-1.90* (1.02)	0.718	0.53 (0.80)	0.668
Last month business profit (USD)	1104	98.60 (143.83)	-6.77 (10.50)	0.586	-12.44 (11.53)	0.718	3.55 (13.67)	0.895
Last month business revenue (USD)	1280	396.21 (625.91)	84.40* (44.75)	0.101	-0.80 (51.71)	0.988	-92.58 (57.75)	0.350
Last month business costs (USD)	1206	276.34 (358.93)	82.04** (36.68)	0.049	25.30 (37.84)	0.775	-98.48** (49.45)	0.350
Last month HH income (USD)	1358	332.33 (473.05)	-34.24 (36.83)	0.425	16.93 (41.60)	0.881	-75.43 (49.79)	0.350

This table shows [Table B5](#) with sharpened FDR q-values following Anderson (2008). Each row is one regression. All variables measuring values are in USD. Results are qualitatively unchanged when using logged versions of continuous outcomes. Sample sizes vary for some questions because of missing data, particularly when respondents were unable to estimate monetary values with a high degree of confidence, or because some questions were only asked to a subset of respondents. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is a household or a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C9: Impact of transformer injection intervention on primary outcomes, accounting for implementation issues and construction timing

	All sites	Comm- issioned sites (SMEC)	New transformer confirmed sites	All sites, IV new tx with treat	No movers	All sites, except close sites
Voltage damage and protection index	-0.10* (0.06)	-0.11* (0.06)	-0.13** (0.07)	-0.14* (0.08)	-0.11* (0.06)	-0.16** (0.08)
Any voltage-related damage, last 12 months (=1)	-0.05 (0.04)	-0.06 (0.04)	-0.07* (0.04)	-0.06 (0.05)	-0.06 (0.04)	-0.08* (0.05)
Any reliability protective device owned (=1)	-0.02 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.02 (0.02)	-0.01 (0.01)	-0.02 (0.02)
Amt. spent on burnt/broken apps in past year (USD)	1.23 (1.90)	1.09 (1.93)	0.21 (2.04)	1.63 (2.53)	0.88 (1.89)	1.49 (2.64)
Value of protective devices owned (USD)	0.32 (1.62)	0.25 (1.65)	1.05 (1.82)	0.43 (2.16)	0.41 (1.72)	-0.85 (1.65)
Reported hours of bad voltage in past month	-9.09 (7.69)	-10.19 (7.80)	-12.79 (7.92)	-12.01 (10.08)	-13.26* (7.16)	-4.85 (8.42)
Reported total outage hours in past month	-1.22 (2.73)	-1.32 (2.76)	-2.01 (2.87)	-1.62 (3.59)	-1.16 (2.84)	1.84 (3.41)
WTP for perfect reliability and quality (USD)	0.44 (0.28)	0.46 (0.29)	0.54* (0.31)	0.58 (0.37)	0.36 (0.30)	0.23 (0.35)
WTP for perfect voltage and half outage hours (USD)	0.21 (0.21)	0.20 (0.21)	0.26 (0.23)	0.27 (0.27)	0.14 (0.21)	-0.00 (0.28)
WTP for no outages and half bad voltage hours (USD)	0.99** (0.44)	1.07** (0.44)	1.14** (0.48)	1.31** (0.58)	0.89** (0.43)	0.70 (0.62)
Total no. of appliances owned	0.09 (0.13)	0.09 (0.13)	0.15 (0.14)	0.12 (0.17)	0.12 (0.13)	0.15 (0.15)
Any alt. energy source used in last month (=1)	-0.00 (0.01)	-0.00 (0.01)	0.00 (0.01)	-0.00 (0.01)	0.00 (0.01)	0.02 (0.02)
Last month electricity spending (USD)	0.53 (0.80)	0.51 (0.81)	0.49 (0.87)	0.70 (1.06)	0.63 (0.83)	0.72 (1.00)
Last month business profit (USD)	3.67 (13.59)	4.16 (13.74)	10.11 (14.69)	4.97 (18.20)	5.20 (14.44)	6.28 (18.27)
Last month business revenue (USD)	-91.60 (57.48)	-90.49 (58.28)	-23.30 (57.11)	-125.79 (82.61)	-110.70* (60.84)	-103.81 (70.22)
Last month business costs (USD)	-97.46** (49.15)	-99.05** (49.64)	-54.44 (49.75)	-129.75* (68.16)	-105.28** (51.70)	-140.71** (59.55)
Last month HH income (USD)	-75.62 (49.63)	-82.31 (50.19)	-85.61 (53.64)	-102.29 (66.69)	-71.11 (52.38)	-53.05 (57.50)

This table shows the same difference-in-difference analyses presented in Table B5. Each cell is one regression. Column 1 replicates the “Post $\times$ Treat” column from Table B5. Column 2 drops two sites in Kaneshie where the construction manager SMEC indicated the new transformer was not commissioned successfully. Column 3 drops sites where our own construction monitoring activities indicated no new transformer was built in a treatment site or a new transformer was built in a control site. Column 4 instruments for observing a new transformer during the construction monitoring visits with site treatment assignment. Column 5 drops respondents who moved between the baseline and endline survey. Column 6 drops control sites that are within 1.3 km from a treatment site (the median distance). Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C10: Correlations between raw voltage quality and primary outcomes

	Full N	Mean (SD)	Average voltage	Hours voltage 20% below nominal
Reported hours of bad voltage in past month	3111	22.72 (60.01)	-0.744*** (0.201)	0.162*** (0.052)
Reported total outage hours in past month	3120	20.52 (33.93)	-0.315*** (0.095)	0.046*** (0.017)
WTP for perfect reliability and quality (USD)	3131	2.51 (3.48)	-0.019*** (0.005)	0.003*** (0.001)
WTP for perfect voltage and half outage hours (USD)	3131	1.32 (2.15)	-0.009*** (0.003)	0.002** (0.001)
WTP for no outages and half bad voltage hours (USD)	847	1.89 (3.13)	-0.005 (0.007)	0.002 (0.002)
Voltage damage and protection index	3131	-0.02 (0.97)	-0.003* (0.002)	0.001** (0.001)
Any voltage-related damage, last 12 months (=1)	3131	0.24 (0.42)	-0.002** (0.001)	0.001** (0.000)
Amt. spent on burnt/broken apps in past year (USD)	3063	6.48 (22.79)	-0.087*** (0.028)	0.018** (0.008)
Amt. spent on burnt/broken apps (if damage = 1)	710	29.95 (47.35)	-0.111 (0.130)	0.006 (0.024)
Any reliability protective device owned (=1)	3131	0.25 (0.43)	-0.000 (0.001)	0.000 (0.000)
Value of protective devices owned (USD)	2649	4.08 (14.50)	-0.031 (0.022)	0.013* (0.007)
Any alt. energy source used in last month (=1)	3131	0.05 (0.21)	0.000 (0.000)	0.000 (0.000)
Total no. of appliances owned	3131	8.47 (5.38)	0.003 (0.006)	0.000 (0.001)
Last month electricity spending (USD)	3031	14.94 (12.92)	0.011 (0.017)	-0.003 (0.003)
Last month business profit (USD)	1103	89.36 (127.40)	0.100 (0.224)	0.019 (0.045)
Last month business revenue (USD)	1273	401.94 (598.45)	2.415*** (0.914)	-0.405* (0.217)
Last month business costs (USD)	1201	294.79 (390.65)	1.496** (0.661)	-0.187 (0.149)
Last month HH income (USD)	1348	301.69 (426.69)	-0.774 (0.782)	0.188 (0.160)

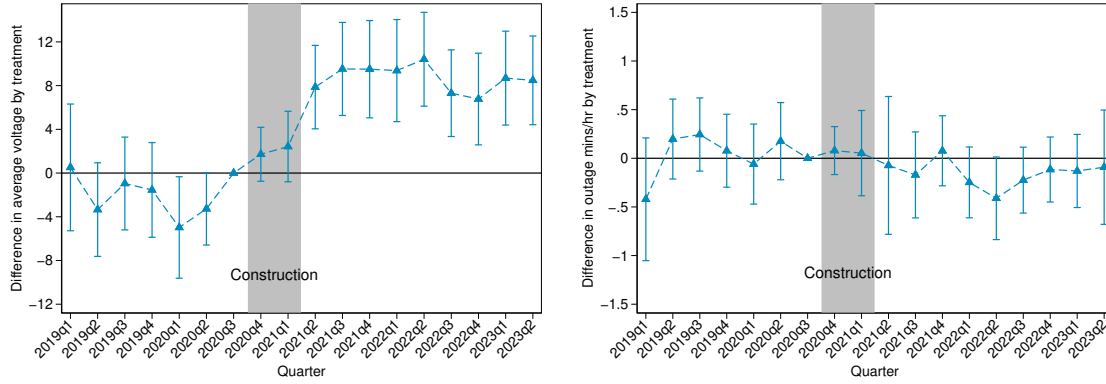
Each cell is one regression. This table shows the results from separate regressions of primary outcomes on measures of voltage quality. This table shows estimates from independent regressions with two voltage measurements as the explanatory variables and each row a different outcome variables, Each row represents a different outcome pooling business and household respondents. Profit is measured by directly asking the respondent, rather than by subtracting costs from revenues. Total reported costs are the sum of costs for specific items/activities and are not comprehensive. The columns indicate measures of voltage quality—the independent variables. Voltage is measured by assigning each respondent GridWatch data based on the nearest devices for the year before the survey date. Mean voltage in control sites is 219.5V at baseline and 224.6V at endline. In all the regressions, we also control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the location includes both a household and a business, and district fixed effects. Standard errors are clustered at the site level. [Table B3](#) shows the same using data adjusted for the device and the respondents' distances to transformer. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C11: Baseline balance by site status – Dropping control sites with high voltage improvement

	Control N	Mean	Treat N	Difference	p-value
<i>Respondent and Location</i>					
Age (years)	619	39.01	803	-0.63	0.305
Respondent is male	619	0.36	803	0.01	0.597
Completed secondary education	619	0.50	803	-0.00	0.931
Owens premises	619	0.36	803	-0.03	0.296
<i>Appliances</i>					
Any television (TV) at location	619	0.68	803	-0.05	0.051
Any fridge at location	619	0.62	803	0.01	0.638
Count of mobiles	619	2.27	803	0.17	0.056
Any reliability protective device owned (=1)	619	0.24	803	-0.03	0.260
Count of reliability protective devices	619	0.34	803	0.00	0.922
<i>Electricity and Energy</i>					
Pays someone else for electricity	619	0.09	803	0.01	0.613
Count of meter users	619	1.77	803	-0.10	0.213
Last month electricity spending (USD)	611	17.96	796	2.15	0.013
Has generator	619	0.05	803	0.01	0.584
Count of alternative fuels used in past 3 months	619	0.89	803	-0.02	0.687
Last month spending on alternative fuels (USD)	619	8.05	803	1.40	0.288
<i>Reliability</i>					
Reported number of outages in past month	619	7.25	803	0.62	0.017
Reported total outage hours in past month	619	41.61	803	1.53	0.562
Reported avg. hours per day with bad voltage	616	1.38	797	-0.34	0.033
Any voltage-related damage, last 12 months (=1)	619	0.25	803	-0.04	0.059
Amt. spent on burnt/broken apps in past year (USD)	616	9.66	794	0.24	0.895
<i>Household Characteristics</i>					
Adult members	284	2.37	383	-0.04	0.691
Child members (<18)	284	1.19	383	-0.04	0.728
Last month HH income (USD)	275	308.62	367	-29.06	0.421
Share of HH adults (18+) with paid jobs in last 7 days	284	0.65	383	-0.03	0.331
<i>Business Characteristics</i>					
Number of workers	335	1.99	420	0.04	0.774
Last month business revenue (USD)	276	388.46	380	-10.01	0.825
Total measured business costs in past month	266	278.14	366	-30.99	0.286
Last month business profit (USD)	250	99.19	336	4.96	0.670
Total hours typically open	335	12.17	420	0.20	0.265
Any non-electric business machines at location	335	0.08	420	-0.01	0.655
Business engaged in retail activities	335	0.44	420	0.00	0.919
Business engaged in manufacturing activities	335	0.21	420	0.02	0.530
Business engaged in other service activities	335	0.35	420	-0.02	0.529
Business activity likely using electricity	335	0.22	420	-0.00	0.970

This table shows means in the baseline period for survey respondents, pooling businesses and households, and tests for significance of the differences in means by line bifurcation treatment status. The control group is restricted to control sites that experienced less than a 10 V improvement in voltage after line-bifurcation, which is the 75th percentile of voltage improvement in control sites. The p-value for the joint F-test for household baseline characteristics is 0.111. The p-value for the joint F-test for business baseline characteristics is 0.136. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure C6: Treatment effect on voltage and outages – Dropping control sites with high voltage improvements



The figure shows the impact of injection transformer intervention on average voltage over time. We report the coefficient estimate in addition to the 95% confidence interval. The control group is restricted to control sites that experienced less than a 10 V improvement in voltage after line-bifurcation, which is the 75th percentile of voltage improvement in control sites. Figure 6 shows the version with the full sample.

Table C12: Impact of transformer injection intervention on outages and voltage – Dropping control sites with high voltage improvements

	(1)	(2)	(3)	(4)	(5)	(6)
	Minutes power out	Average voltage	Absolute voltage deviation	Standard deviation voltage	Hours of spells >10% below nominal	Hours of spells >20% below nominal
During construction	0.16* (0.08)	-1.30 (1.00)	0.89 (0.91)	0.15*** (0.05)	16.00 (11.97)	11.43 (10.38)
Treat X During	0.01 (0.12)	4.32*** (1.54)	-3.01** (1.29)	-0.33*** (0.10)	-52.93*** (18.37)	-44.28*** (15.61)
Post	-0.07 (0.09)	0.61 (0.89)	-0.60 (0.65)	0.06 (0.06)	2.39 (9.61)	2.12 (7.44)
Treat X Post	-0.21 (0.14)	10.76*** (1.99)	-3.96*** (1.42)	-0.75*** (0.10)	-100.42*** (18.09)	-76.00*** (14.52)
Observations	8581627	8435976	8435976	8406529	12183	12183
Pre-constr. ctl. mean	1.39	222.28	14.07	2.29	93.30	50.89
Hour of day FE	Y	Y	Y	Y	N	N
Week of year FE	Y	Y	Y	Y	N	N
Month of year FE	N	N	N	N	Y	Y
Site FE	Y	Y	Y	Y	Y	Y
Hourly/monthly data	Hourly	Hourly	Hourly	Hourly	Monthly	Monthly

Difference-in-difference results for the impact of treatment on power quality measured by GridWatch devices, with each column a different outcome variable. The control group is restricted to control sites that experienced less than a 10 V improvement in voltage after line-bifurcation, which is the 75th percentile of voltage improvement in control sites. Columns (1)-(4) use hourly data while Columns (5)-(6) use monthly data. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C13: Impact of transformer injection intervention on customer electricity experience – Dropping control sites with high voltage improvements

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post $\times$ Treat (SE)
Voltage damage and protection index	2844	-0.02 (0.99)	-0.11** (0.05)	0.10 (0.07)	-0.11 (0.07)
Any voltage-related damage, last 12 months (=1)	2844	0.25 (0.43)	-0.05 (0.03)	0.04 (0.03)	-0.05 (0.04)
Any reliability protective device owned (=1)	2844	0.24 (0.43)	-0.02** (0.01)	0.02 (0.02)	-0.02* (0.01)
Amt. spent on burnt/broken apps in past year (USD)	2784	9.78 (35.31)	-6.37*** (1.54)	-0.61 (1.93)	1.81 (2.04)
Value of protective devices owned (USD)	2412	5.79 (25.04)	-2.50** (1.10)	-0.13 (1.51)	0.51 (1.72)
Reported hours of bad voltage in past month	2806	41.59 (85.20)	-40.94*** (5.18)	8.27 (7.73)	-10.69 (7.98)
Reported total outage hours in past month	2778	33.98 (31.89)	-31.15*** (2.33)	-0.38 (2.81)	0.66 (2.96)
WTP for perfect reliability and quality (USD)	2844	3.23 (4.38)	-1.33*** (0.22)	-0.30 (0.26)	0.35 (0.29)
WTP for perfect voltage and half outage hours (USD)	2844	1.51 (2.58)	-0.28* (0.16)	-0.10 (0.18)	0.11 (0.21)
WTP for no outages and half bad voltage hours (USD)	756	1.69 (2.80)	0.20 (0.28)	-0.16 (0.32)	0.72* (0.43)
Total no. of appliances owned	2844	8.66 (6.26)	-0.03 (0.08)	-0.29 (0.37)	0.11 (0.13)
Any alt. energy source used in last month (=1)	2844	0.05 (0.23)	-0.01 (0.01)	0.00 (0.01)	-0.00 (0.01)
Last month electricity spending (USD)	2750	18.11 (17.20)	-4.42*** (0.70)	-2.23** (1.13)	1.03 (0.87)
Last month business profit (USD)	996	93.08 (139.22)	3.44 (10.51)	-6.23 (11.74)	-6.81 (13.68)
Last month business revenue (USD)	1156	389.22 (577.33)	114.26** (51.83)	11.24 (46.59)	-122.24* (63.08)
Last month business costs (USD)	1096	276.01 (334.77)	87.38** (43.53)	36.19 (34.95)	-103.92* (54.63)
Last month HH income (USD)	1216	309.08 (416.50)	3.68 (37.92)	37.98 (40.66)	-113.41** (50.64)

This table shows the difference-in-difference results from the [Equation 2](#) pooling businesses and households. Each row presents results from one regression with a different socio-economic variable as the outcome. The control group is restricted to control sites that experienced less than a 10 V improvement in voltage after line-bifurcation, which is the 75th percentile of voltage improvement in control sites. All outcomes pre-specified in the pre-analysis plan (Berkouwer et al., 2019), except for voltage improvements as these were unanticipated.. All variables measuring values are in USD. Results are qualitatively unchanged when using logged versions of continuous outcomes. Sample sizes vary for some questions because of missing data, particularly when respondents were unable to estimate monetary values with a high degree of confidence, or because some questions were only asked to a subset of respondents. Reliability outcomes are measured using respondent self-reports based on the 30 days prior to the survey date at both baseline and endline. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is a household or a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C14: Impact of transformer injection intervention on main business outcomes – Dropping control sites with high voltage improvements

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post $\times$ Treat (SE)
Last month business profit (USD)	996	93.08 (139.22)	3.44 (10.51)	-6.23 (11.74)	-6.81 (13.68)
Last month business costs (USD)	1096	276.01 (334.77)	87.38** (43.53)	36.19 (34.95)	-103.92* (54.63)
Last month wage and benefits costs (USD)	1208	49.51 (127.31)	18.61* (9.63)	0.56 (9.16)	-9.41 (12.26)
Last month materials costs (USD)	1150	192.00 (290.79)	73.85** (34.82)	47.87* (28.42)	-94.92** (44.33)
Last month electricity spending (USD)	1446	17.78 (17.36)	-4.16*** (0.79)	-1.79 (1.29)	0.47 (1.01)
Last month spending on alternative fuels (USD)	1510	5.61 (41.22)	-1.51 (2.12)	-1.91 (2.44)	1.72 (2.25)
Last month business revenue (USD)	1156	389.22 (577.33)	114.26** (51.83)	11.24 (46.59)	-122.24* (63.08)
Estimated increase in revenue w/ perfect electricity (USD)	950	434.91 (936.36)	-219.23*** (82.65)	5.29 (83.64)	-105.57 (112.80)
Number of workers	1510	1.99 (1.98)	0.09 (0.07)	-0.08 (0.15)	0.09 (0.09)
Share of men employees	1500	0.31 (0.41)	-0.01 (0.01)	0.00 (0.01)	-0.00 (0.02)
Share of full-time employees	1484	0.90 (0.22)	-0.05*** (0.02)	0.01 (0.02)	0.00 (0.02)
Business open during any 'dark' hours	1510	0.78 (0.41)	-0.11*** (0.03)	-0.03 (0.03)	0.01 (0.04)
Total hours typically open	1510	12.17 (2.45)	-0.60*** (0.15)	-0.18 (0.20)	-0.14 (0.23)
Applied for loans in past 12 months	1510	0.16 (0.37)	0.01 (0.03)	0.06* (0.03)	-0.03 (0.04)
Total value of outstanding loans (USD)	1454	345.45 (1203.81)	13.56 (83.16)	103.11 (93.59)	-204.51* (114.57)

This table shows the difference-in-difference results from the main equation. Each row presents results from one regression with a different socio-economic variable as the outcome. All variables measuring values are in USD. Profit is measured by directly asking the respondent, rather than by subtracting costs from revenues. Total reported costs are the sum of costs for specific items/activities and are not comprehensive. The control group is restricted to control sites that experienced less than a 10 V improvement in voltage after line-bifurcation, which is the 75th percentile of voltage improvement in control sites. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is the business owner or a manager, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Appendix D: Pre-specified analyses of socioeconomic outcomes

Table D1: Impact of transformer injection intervention on willingness to pay outcomes

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post x Treat (SE)
WTP for perfect reliability and quality (USD)	3150	3.29 [4.41]	-1.41*** (0.21)	-0.34 (0.25)	0.44 (0.28)
WTP for perfect voltage and half outage hours (USD)	3150	1.58 [2.71]	-0.37** (0.16)	-0.17 (0.17)	0.21 (0.21)
Max monthly WTP for 1 unnan. 8hr outage	240	0.97 [1.72]	-0.51* (0.27)	-0.07 (0.30)	-0.17 (0.35)
Max monthly WTP for 1 announ. 8hr outage	262	1.27 [2.08]	-0.77*** (0.28)	0.10 (0.44)	0.12 (0.48)
Max monthly WTP for 4 unnan. 2hr outages	244	1.38 [2.20]	-0.84** (0.34)	-0.39 (0.40)	0.25 (0.43)
WTP for no outages and half bad voltage hours (USD)	850	1.95 [3.34]	-0.07 (0.30)	-0.37 (0.32)	0.99** (0.44)
Max WTP for generator	2724	317.00 [403.13]	-14.43 (24.44)	33.76 (23.80)	-38.35 (31.95)

Additional results from [Equation 2](#). All variables measuring values are in USD. Results are qualitatively unchanged when using logs. Sample sizes are lower for reliability scenarios that were only presented to a random subset of respondents. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D2: Impact of transformer injection intervention on electricity-related indices

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post x Treat (SE)
Outage backup power index	3150	0.00 [1.00]	-0.06 (0.04)	0.02 (0.06)	-0.01 (0.04)
Freq. of wetcell batt./generator use during outage (normalized)	3150	0.06 [1.26]	-0.06 (0.04)	-0.00 (0.07)	0.02 (0.05)
Share of apps. using solar/generator during outage (normalized)	3114	0.10 [1.41]	-0.10 (0.06)	0.04 (0.09)	-0.04 (0.08)
Alternative energy/fuel sources index	3150	-0.00 [1.00]	-0.02 (0.05)	0.03 (0.07)	-0.03 (0.07)
Any alt. energy source used in last month (=1)	3150	0.05 [0.22]	-0.01 (0.01)	0.00 (0.01)	-0.00 (0.01)
Count of alt. light sources (normalized)	3150	0.10 [1.10]	-0.10 (0.06)	0.05 (0.09)	-0.05 (0.09)
Count of alt. fuel sources (normalized)	3150	-0.08 [0.96]	0.08** (0.03)	-0.00 (0.05)	0.01 (0.05)
Appliance protection index	3150	-0.00 [1.00]	-0.18*** (0.03)	-0.04 (0.05)	0.07* (0.04)
Count of reliability defensive apps. (normalized)	3150	0.08 [1.11]	-0.08*** (0.02)	-0.02 (0.06)	0.02 (0.03)
Has multi-phase system	2516	0.04 [0.21]	0.01* (0.01)	0.01 (0.01)	-0.01 (0.01)
Share of TVs plugged to TV guard (normalized)	110	2.54 [0.77]	-1.31*** (0.27)	-0.36 (0.24)	0.16 (0.39)
Share of fridges plugged to fridge guard (normalized)	232	1.35 [0.45]	-0.47*** (0.14)	-0.09 (0.10)	0.16 (0.20)

Additional results from [Equation 2](#). The main outcomes are indices; we also show results for the index components for completeness. Indices are constructed as the sum of normalized components, and are then normalized to have mean 0 and SD 1 for control respondents in the baseline. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D3: Impact of transformer injection intervention on additional business outcomes

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post x Treat (SE)
Temporary business response type index	1658	0.00 [1.00]	-0.10 (0.06)	0.03 (0.09)	0.07 (0.10)
Temporary switch to alternative energy due to reliability (0-2)	1646	0.06 [0.29]	0.04*** (0.02)	0.02 (0.02)	-0.01 (0.03)
Temporary stop working, work less due to reliability (0-2)	1646	0.42 [0.71]	-0.03 (0.04)	0.02 (0.06)	-0.01 (0.06)
Temporary postpone working, work same due to reliability (0-2)	1646	0.33 [0.63]	-0.13*** (0.03)	-0.02 (0.05)	-0.01 (0.05)
Temporary switch tools/labor due to reliability (0-2)	1646	0.16 [0.44]	0.01 (0.03)	-0.01 (0.03)	0.03 (0.04)
Temporary switch business activities due to reliability (0-2)	1646	0.15 [0.41]	-0.03 (0.03)	0.02 (0.03)	0.04 (0.04)
Temporary reduce labor due to reliability (0-2)	1646	0.20 [0.49]	-0.05 (0.04)	0.01 (0.04)	0.02 (0.05)
Other temporary response due to reliability	1646	0.02 [0.17]	-0.02* (0.01)	0.00 (0.01)	0.02 (0.02)
Temporary business response intensity index	1658	-0.00 [1.00]	-0.33*** (0.05)	0.61 (0.41)	-0.61 (0.41)
Days of switching to solar energy	1658	0.00 [0.00]	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Days of switching to generator	1658	0.01 [0.07]	-0.01* (0.00)	0.04 (0.04)	-0.05 (0.04)
Days of switching to wetcell	1658	0.00 [0.01]	-0.00 (0.00)	0.01 (0.01)	-0.01 (0.01)
Days stopping or postponing work in past 1 month	1658	0.00 [1.00]	-0.32*** (0.05)	0.02 (0.07)	-0.01 (0.07)
Percentage of business hours stopping work	1658	-0.00 [1.00]	-0.34*** (0.06)	0.09 (0.09)	-0.07 (0.09)
Permanent business response index	1658	0.00 [1.00]	-0.11 (0.07)	-0.09 (0.07)	0.27** (0.12)
Permanently substitute to non-electric tools or machines (0-1)	1646	0.02 [0.13]	-0.01 (0.01)	-0.01** (0.01)	0.01 (0.01)
Permanently substitute to more labor (0-1)	1646	0.01 [0.09]	-0.00 (0.00)	0.00 (0.01)	0.02 (0.01)
Permanently change industry/business (0-1)	1646	0.01 [0.12]	-0.00 (0.01)	-0.01 (0.01)	0.03** (0.01)
Purchase generator (0-1)	1646	0.01 [0.12]	-0.01 (0.01)	-0.01 (0.01)	0.01 (0.01)
Business qualitative assessments index	1658	0.00 [1.00]	-0.76*** (0.07)	0.00 (0.08)	-0.06 (0.09)
Perceived safety in area (1-5)	1656	3.30 [1.05]	0.00 (0.08)	-0.06 (0.10)	0.16 (0.11)
Importance of electricity as obstacle to business (1-5)	1646	3.90 [0.98]	-0.23*** (0.08)	0.01 (0.08)	-0.06 (0.11)
Belief that Dumsor is back (1-5)	1646	2.84 [1.31]	1.50*** (0.10)	0.14 (0.11)	-0.06 (0.12)
Expected reliability one year from today (1-3)	1078	2.34 [0.83]	0.35*** (0.08)	-0.00 (0.09)	0.05 (0.10)
Importance of finance/access to credit as a business obstacle (1-5)	1658	2.79 [1.29]	-0.37*** (0.11)	0.02 (0.09)	0.11 (0.15)

All variables measuring values are in USD. Results are qualitatively unchanged when using logs. Indices are constructed as the sum of normalized components, and are then normalized to have mean 0 and SD 1 for control respondents in the baseline. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$